

Predicting Protected Area Designations and Assessing Transition Risk under 30×30

Master Thesis

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Abstract

The Kunming-Montreal Global Biodiversity Framework commits nearly 200 countries to protect 30% of terrestrial land by 2030, up from approximately 17% today. Where the additional designations will occur is unknown in advance yet materially consequential: protected-area creation imposes binding land-use restrictions that can strand agricultural, mining, and infrastructure assets, generating *transition risk* for landowners, investors, and financial supervisors. This thesis develops a machine-learning framework that quantifies this transition risk at the pixel level by predicting the probability that any 1 km² parcel becomes newly designated as a protected area within a five-year horizon. The framework is applied to South America, the United States, and Southeast Asia, training LightGBM and Random Forest models on a pixel-year panel of historical designation patterns covering 2000–2024. In South America, the framework achieves strong out-of-sample discrimination (ROC-AUC 0.897; Precision–Recall AUC 0.645): the top 1% of predicted pixels capture 90.4% of all observed protected-area establishments in the held-out test period, confirming that calibrated probability maps identify actionable transition-risk concentrations. Feature importance analysis reveals that historical PA designation is jointly driven by ecological conservation priority and institutional capacity, not purely by ecological suitability; high-value agricultural land is systematically suppressed from high-risk zones, consistent with an opportunity-cost mechanism. Forward inference projects that under business-as-usual designation rates all three regions fall far short of the 30×30 target. Cross-continental transfer experiments show that predictive patterns are region-specific, implying that the expansion pathway is shaped by regional political economies rather than universal ecological gradients. The results provide a spatially explicit transition-risk tool for investors, financial supervisors, and policymakers anticipating the next wave of protected-area expansion.

Keywords: protected areas; 30×30; transition risk; machine learning; LightGBM; spatial prediction; land-use change; biodiversity policy

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List of Abbreviations

Abbreviation	Definition
BAU	Business-as-usual
CBD	Convention on Biological Diversity
DPI	Database of Political Institutions
ECE	Expected Calibration Error
GBM	Gradient-Boosting Machine
GSN	Global Safety Net
HNTL	Harmonised Night-Time Lights
LOBO	Leave-One-Biome-Out
MCE	Maximum Calibration Error
MODIS	Moderate Resolution Imaging Spectroradiometer
NDVI	Normalised Difference Vegetation Index
OECM	Other Effective Area-Based Conservation Measure
PA	Protected Area
PR-AUC	Precision–Recall Area Under the Curve
RF	Random Forest
ROC-AUC	Receiver Operating Characteristic Area Under the Curve
SHAP	SHapley Additive exPlanations
SRTM	Shuttle Radar Topography Mission
V-Dem	Varieties of Democracy
WDPA	World Database on Protected Areas
WGI	Worldwide Governance Indicators
WRI	World Resources Institute

1 Introduction

The global expansion of protected areas (PAs) is a central pillar of contemporary biodiversity policy. The Kunming-Montreal Global Biodiversity Framework, adopted at COP15 in December 2022, commits nearly 200 signatory countries to designating 30% of the world’s land and oceans as protected by 2030: the so-called 30×30 target (Convention on Biological Diversity, 2022). At the time of writing, terrestrial protection stands at approximately 17% (UNEP-WCMC and IUCN, 2026), meaning governments must nearly double the protected land area in under five years. While this commitment is widely regarded as necessary to halt biodiversity loss, large-scale PA expansion carries significant economic and regulatory consequences. PA designation typically introduces binding land-use restrictions that constrain agriculture, resource extraction, infrastructure development, and urban expansion (Schuster et al., 2023). Uncertainty about where new PAs will be established therefore generates material *transition risk* for businesses, investors, and policymakers: the risk that future conservation decisions strand assets or alter the viability of existing land uses (Govind and Webb, 2025).

The economic stakes are substantial. Protected-area policies impose real opportunity costs in the form of foregone rents, constrained resource extraction, and reduced land values (Reynaert et al., 2024). These costs are spatially heterogeneous: they fall disproportionately on landowners, farmers, and investors whose assets sit in locations most likely to be designated (Robinson et al., 2024). The concern is not limited to direct asset-owners. Financial supervisors and central banks have increasingly identified nature-related risks, including abrupt land-use transitions, as potential sources of systemic financial exposure (Govind and Webb, 2025). A spatially explicit PA probability model directly addresses these supervisory needs: knowing *where* designation pressure is concentrated allows banks and asset managers to quantify transition risk in their portfolios, stress-test collateral values, and communicate material nature-related risk exposures to regulators. Equally, better-informed transition-risk expectations can reduce the uncertainty premium that landowners demand as compensation for designation risk, thereby lowering resistance to conservation policy and accelerating the expansion that the 30×30 target requires. Nature-related financial disclosure frameworks

explicitly identify PA creation as a transition risk category affecting business operations, supply chains, and land-secured lending (Govind and Webb, 2025).

Critically, historical PA designation is not spatially random. The empirical literature in section 2.2 documents that governments systematically protect land that is remote, ecologically important, and economically cheap to designate. These areas are characterised by high elevation, steep slopes, low population density, and distance from infrastructure such as roads and power plants. (Joppa and Pfaff, 2009; Venter et al., 2018; Baldi et al., 2017; Mouillot et al., 2024). This bias reflects the political economy of conservation: designation is easiest where opportunity costs are lowest. If past patterns predict future patterns, then geographic covariates can be used to build a forward-looking transition-risk model. Recent work confirms this principle (Mouillot et al., 2024; Greenhill et al., 2024), but existing approaches remain limited: they are typically designed to map ecological suitability rather than to generate calibrated probability forecasts for government designation behaviour, they focus on single regions, and they do not produce the probability estimates required for quantitative risk assessment.

This thesis addresses these gaps by developing a multi-region machine-learning framework that targets the prediction of future PA establishment. The framework treats PA designation as a rare binary transition in a pixel-year panel, trains LightGBM and Random Forest models on historical designation patterns, calibrates the raw scores to interpretable probabilities, and deploys the calibrated model to generate forward-looking transition-risk maps for the 2025–2030 period. Three regions are studied: South America (the primary model), and the United States and Southeast Asia as robustness checks providing cross-context validation. The analysis produces four concrete outputs: (i) calibrated spatial risk scores that quantify each pixel’s five-year designation probability; (ii) forward probability maps and on-track assessments under business-as-usual, moderate, and 30×30 scenarios; (iii) estimates of economic exposure (measured by agricultural land area) in the highest-risk zones, directly relevant for portfolio risk assessment; and (iv) feature importance analysis that identifies the ecological, institutional, and economic drivers of government designation decisions, revealing how closely the projected expansion tracks genuine conservation priority.

The main results are as follows. In South America, LightGBM achieves an out-of-sample

ROC-AUC of 0.897 and a Precision–Recall AUC of 0.645. The top 1% of predicted pixels (approximately 208,000 km²) captures 90.4% of all actual PA designations in the held-out test period, at a lift of 66.8× over random. Feature importance analysis reveals that designation follows three reinforcing channels: ecological conservation priority, institutional feasibility, and low economic cost; high-value agricultural land is systematically absent from high-risk zones. Forward projections under business-as-usual designation rates indicate that South America would reach the 30% target only after 67 additional years, with the United States and Southeast Asia falling even further short. Cross-continental transfer experiments show that predictive patterns learned in one region do not transport to others, confirming that designation dynamics reflect region-specific political economies rather than universal ecological gradients. This region-specificity is a design feature, not a limitation: each regional model captures the governance context that actually shapes designation, making the resulting risk maps operationally useful for portfolio-level exposure analysis.

The remainder of the thesis is organised as follows. Section 2 reviews the four strands of literature this work connects. Section 3 describes the data sources, study areas, and panel construction. Section 4 presents the modelling framework and evaluation strategy. Section 5 presents the results for all three regions. Section 6 discusses the implications for conservation policy, investor risk assessment, and the limitations of the approach. Section 7 concludes.

2 Literature Review

2.1 Biodiversity and Conservation Policy

Global biodiversity is declining at a rate unprecedented in human history (Díaz et al., 2019). The IPBES Global Assessment documents that human activities have already substantially altered three-quarters of the Earth’s land surface and that one million plant and animal species face extinction (Díaz et al., 2019). The drivers, including land-use change, over-exploitation, climate change, invasive species, and pollution, compound ecological pressure and degrade the capacity of ecosystems to sustain human well-being. The loss is not only ecological: biodiversity underpins food security, freshwater provision, disease regulation, and

climate stabilisation, among many other ecosystem services (Millennium Ecosystem Assessment, 2005).

Area-based conservation, defined as the formal legal designation of land as protected from extractive and destructive uses, is the primary policy instrument deployed in response (Maxwell et al., 2020). Maxwell et al. (2020) and Watson et al. (2014) synthesise evidence from across the global PA network and show that, where adequately managed, protected areas measurably reduce habitat loss and maintain higher biodiversity than surrounding unprotected landscapes. However, the same reviews identify a persistent gap between the formal extent of protection and its functional effectiveness: many PAs are underfunded, cover ecologically marginal land, or lack the management capacity to deter illegal activities. The world’s most threatened species and ecosystems frequently remain outside the protected network (Williams et al., 2022; Chaplin-Kramer et al., 2023).

These shortcomings provided the impetus for the Kunming-Montreal Global Biodiversity Framework, adopted at COP15 in December 2022, which commits nearly 200 countries to protecting 30% of terrestrial and marine areas by 2030 (Convention on Biological Diversity, 2022). At the time of writing, terrestrial coverage stands at approximately 17% (UNEP-WCMC and IUCN, 2026), meaning governments must designate approximately the same land area as the entire existing global PA network within five years. The pace and spatial pattern of this expansion will largely determine whether the 30×30 commitment delivers meaningful conservation outcomes or, as past trends warn, concentrates new designations in ecologically marginal areas (Joppa and Pfaff, 2009; Venter et al., 2018).

The biodiversity and climate crises are increasingly recognised as intertwined policy problems. Carbon-dense ecosystems such as tropical forests, peatlands, and wetlands are simultaneously among the world’s most species-rich environments (Dinerstein et al., 2020), so protecting them serves dual objectives: conserving ecological communities and preserving terrestrial carbon stocks critical for meeting Paris Agreement targets (IPCC, 2019). The Global Safety Net (Dinerstein et al., 2020) explicitly maps “climate stabilisation areas” as a core component of the global conservation footprint, reflecting this convergence. In practice, the overlap creates financial incentives for PA designation: payments for avoided deforestation and voluntary carbon-market schemes make protecting high-carbon-stock forests

economically attractive even where purely ecological arguments would not overcome the opportunity cost of designation (Busch et al., 2022). This dual motivation is especially relevant in the tropical study regions (South America and Southeast Asia), where forest conservation generates both carbon credits and biodiversity benefits. From a risk-management perspective, the intersection reinforces the transition-risk framing: landowners and investors in carbon-rich landscapes face compounding regulatory pressure as climate and biodiversity commitments converge (Boran and Pettorelli, 2024; Govind and Webb, 2025).

2.2 Biases in Protected-Area Siting

A foundational empirical regularity found in conservation literature is that governments do not designate PAs at random, nor do they reliably prioritise the land most in need of protection (Joppa and Pfaff, 2009; Venter et al., 2018; Baldi et al., 2017). Joppa and Pfaff (2009) provide a global analysis: across a large sample of countries, PAs are systematically concentrated at high elevations, on steep slopes, in remote locations far from roads and settlements, and on land with low agricultural productivity. These characteristics are precisely those that minimise the opportunity cost of designation, suggesting that governments protect land where protection is politically and economically easy rather than where it is ecologically most urgent (Venter et al., 2018; Mouillot et al., 2024).

Venter et al. (2018) extend this analysis across multiple decades and show that the bias has been remarkably persistent, with PA systems continuing to expand preferentially into economically marginal land even as global conservation commitments have strengthened. Baldi et al. (2017) characterise this pattern as “opportunity-driven” siting: designation follows the path of least resistance. Sharma et al. (2025) further demonstrate that privately managed protected areas exhibit even greater bias toward low-productivity land than publicly managed PAs, indicating that the opportunity-cost logic operates across governance types.

The consequence of this pattern is a gap between where protection is concentrated and where conservation value is highest. Williams et al. (2022) find that the current global PA network is insufficient to safeguard half the world’s mammal species from extinction because their ranges fall disproportionately outside the protected network. Chaplin-Kramer et al.

(2023) show that only 15% of the “critical natural assets”, defined as ecosystems providing 90% of nature’s contributions to people globally, from freshwater to carbon storage, currently lie within formal protected areas.

Mouillot et al. (2024) synthesise these findings using a machine-learning approach, formalising a “socio-environmental niche” of protected areas: the multivariate space of geographic, climatic, and socioeconomic conditions characteristic of protected relative to unprotected land. Their results confirm that a compact set of covariates, including remoteness, low agricultural suitability, proximity to existing PAs, and national income, explains a large share of the global distribution of PAs. This provides the empirical basis for the approach taken here: systematic, learnable siting patterns in covariate space make it possible to train a supervised model on historical designations and forecast where future ones will occur.

2.3 Machine Learning for Conservation Policy

Machine learning has become a standard toolkit in quantitative conservation science (Pichler and Hartig, 2023). Most applications target biophysical outcomes: habitat suitability mapping from species-occurrence data, land-cover change detection from remote-sensing imagery, and deforestation-risk modelling from geospatial predictors (Ball et al., 2022). Mayfield et al. (2017) provide an influential demonstration that Random Forest classifiers (Breiman, 2001) trained on freely available satellite and topographic predictors match the accuracy of purpose-built deforestation models in Mexico and Madagascar. Their finding that publicly available data are sufficient for high-quality land-change prediction is directly relevant here, as the predictor set of this thesis relies entirely on such sources.

A methodologically distinct strand applies machine learning to *policy-driven* land-use decisions, asking not which habitats are ecologically suitable but which boundaries or designations governments actually impose. Greenhill et al. (2024) train a classifier on geospatial covariates to predict which rivers and wetlands fall under federal regulatory jurisdiction under the US Clean Water Act; their model achieves high accuracy and demonstrates that machine learning can reverse-engineer the implicit spatial decision rules of regulatory agencies from observed outcomes alone. The task is methodologically analogous to predicting where governments will establish protected areas. Kleinberg et al. (2018) provide the theo-

retical underpinning: in many institutional prediction problems, models trained on historical decisions substantially outperform unaided human judgment by exploiting high-dimensional covariate patterns that are too complex for human experts to track consistently.

These studies jointly establish three conditions that motivate the approach of this thesis. First, government conservation decisions follow systematic spatial patterns in geographic covariate space. Second, those patterns are recoverable from historical designation records using standard geospatial predictors. Third, gradient-boosted tree models (Ke et al., 2017) are well suited for this task: they handle non-linear interactions among covariates without requiring prior specification of functional forms, naturally manage the extreme class imbalance of rare transition events, and support post-hoc interpretation via SHapley Additive exPlanations (SHAP) values (Lundberg and Lee, 2017).

Despite this progress, existing machine-learning applications in conservation do not target the specific problem of predicting near-term PA designation. Most models are retrospective or descriptive, classifying existing land cover or regulatory status, or focus on ecological rather than institutional outcomes. No prior study combines a temporal panel structure, multi-region comparison, probability calibration, and forward deployment into a coherent tool for PA transition-risk assessment. This thesis fills that gap.

2.4 30×30 and Transition Risk

Reaching the 30×30 target by 2030 requires an unprecedented acceleration of PA establishment. Robinson et al. (2024) document that achieving 30% protection will require not only expanding conventional government-managed PAs but also incorporating “Nature’s Strongholds” as large, intact wilderness areas alongside novel governance modalities such as indigenous and community conserved areas and other effective area-based conservation measures (OECMs). Dasgupta et al. (2025) analyse implementation experiences across countries and find that designation pace is highly heterogeneous and constrained by land availability, political will, and financial resources; under current trajectories, most countries fall far short of the target.

The economic consequences of large-scale PA expansion are significant. PA designation imposes binding land-use restrictions that curtail agriculture, resource extraction, and

infrastructure development (Schuster et al., 2023). Reynaert et al. (2024) quantify the environmental and economic effects of PA policy and document substantial land-value and income effects for affected communities and landowners. These costs are spatially concentrated: they fall disproportionately on those whose assets sit in the predicted expansion zone. The inability to anticipate where the next wave of PAs will be established therefore generates material uncertainty for landowners, farmers, agribusinesses, and investors with land-linked exposures (Govind and Webb, 2025).

Emerging nature-related financial disclosure frameworks have begun to classify this uncertainty as a formal category of *transition risk* (Govind and Webb, 2025; Taskforce on Nature-related Financial Disclosures, 2023), analogous to the stranded-asset risk created by carbon pricing in the climate-finance literature (Govind and Webb, 2025). Institutions with land-collateralised lending books, agricultural supply chains, or investments in resource-extraction concessions all face potential exposure to PA designation, particularly as political urgency around 30×30 mounts (Taskforce on Nature-related Financial Disclosures, 2023). Despite this growing demand for spatially explicit transition-risk inputs, no prior academic study provides calibrated, forward-looking probability maps for near-term PA establishment that could be directly used in portfolio risk assessment. This thesis addresses that gap: by training a temporal machine-learning model on historical PA designation patterns, calibrating its outputs to interpretable probabilities, and deploying it forward to the 2025–2030 horizon, it provides a spatial risk tool directly relevant to nature-related financial disclosure.

3 Data

3.1 Study Areas

South America. South America is selected as the primary study region for three reasons. First, it contains some of the world’s most biodiverse and ecologically heterogeneous ecosystems (Díaz et al., 2019; Mouillot et al., 2024), including the Amazon rainforest, the Cerrado savanna, the Andean highlands, and the Atlantic Forest, providing a diverse spatial canvas on which to train and evaluate the model. Second, protected-area coverage has expanded

substantially since 2000, expanding from roughly 15% to 25.6% of terrestrial area (UNEP-WCMC and IUCN, 2026), generating sufficient temporal variation in designations to train a transition model. Third, the continent encompasses wide variation in institutional capacity, governance quality, and economic development (World Bank, 2025; V-Dem Institute, 2025), enabling the model to capture a broad range of drivers.

The study covers all terrestrial land areas of South America at a 1×1 km resolution, yielding approximately 20.8 million unique grid cells observed annually from 2000 to 2024 (25 time periods), for a total of approximately 413 million pixel-year observations after excluding already-protected pixels. Processing this dataset at continental scale, spanning three regions totalling over 700 million pixel-year observations, required memory-efficient streaming I/O based on Apache Parquet partitioning and parallelised SLURM array jobs on the ETH Zurich Euler high-performance computing cluster.

United States The United States is chosen as a robustness check because it provides a structurally distinct test environment: a federal system with transparent public-land designations, high data quality, and a current PA coverage of approximately 7.7%. While the US signed the Kunming-Montreal Global Biodiversity Framework, domestic implementation operates through executive orders and congressional mandates rather than binding international commitments, making the intensity of future designation pressure more uncertain than in South America. This institutional contrast is precisely the point: the US serves as a robustness test of whether the predictive framework, trained on a different governance structure and designation process, still recovers historically coherent risk patterns. Substantial federal land holdings (Bureau of Land Management, National Forest) provide a large pool of candidate parcels, and the observed low designation rate creates a severe class-imbalance environment that stresses the model’s calibration and rare-event detection capabilities.

Southeast Asia Southeast Asia is chosen as a second robustness region because its biome composition, including tropical and subtropical forests, mangroves, and montane ecosystems, partly mirrors South America’s, making it a natural candidate to test whether predictive patterns learned on one forest-dominated region transfer to another. At the same time,

the region’s governance heterogeneity (centralized states, archipelagic federations, and post-colonial conservation institutions across ASEAN member states) and its current PA coverage of 14.0% provide a challenging out-of-sample environment. The combination of high deforestation pressure and well-documented PA network expansion over the study period ensures a sufficient signal in the transition target, while the distinct political context relative to South America allows a meaningful test of whether designation dynamics are truly regional.

3.2 Variables and Data Sources

3.2.1 Target Variable

Protected-area boundaries and designation dates are obtained from the World Database on Protected Areas (WDPA; UNEP-WCMC and IUCN, 2024, 2026), the most comprehensive global PA inventory. WDPA polygons are rasterised to the 1 km reference grid for each year of the panel: a cell is classified as protected in year t if any portion of the cell overlaps with a WDPA polygon whose designation year is $\leq t$. This cumulative definition ensures that once protected, a cell remains so, consistent with the transition formulation in Equation (1).

The WDPA records the initial transition into formal protected status. If a PA is subsequently de-listed or has its boundaries contracted, a rare event in the data, the cell retains protected status in our representation. This choice is conservative in the sense that it records more protection, but it reflects the reality that PA de-listing is unusual and that transition-risk analysis is concerned with the risk of designation, not of de-listing.

A known limitation of the WDPA is that designation years can be imprecise: gazette dates may lag the formal legal decision by one to three years in some jurisdictions (Bingham et al., 2019), or may be retrospectively entered into the database. This uncertainty is mitigated by using a five-year lookahead window, which reduces sensitivity to exact label dates.

3.2.2 Predictor Variables

Predictor variables are assembled to capture three broad categories of influence on PA designation: (i) existing conservation context, (ii) environmental and ecological conditions, and (iii) human pressure and economic activity. Table 1 summarises the full set of predictors,

their sources, temporal coverage, and policy context.

Table 1: Predictor variables and data sources used in all three regional models.

Variable / Group	Description	Source & Coverage
Existing Conservation Context		
Distance to nearest PA	Euclidean distance to the boundary of any existing PA in year $t - 1$.	WDPA (UNEP-WCMC and IUCN, 2024); annual 2000–2024
Environmental and Ecological Conditions		
Vegetation index (NDVI)	Annual median Normalised Difference Vegetation Index; proxy for vegetation density and ecosystem productivity.	MODIS MOD13Q1 (Didan, 2021); annual 2000–2024
Deforestation / forest cover	Tree canopy cover (2000 baseline) and cumulative annual forest loss; captures habitat extent and land-use pressure.	Hansen Global Forest Change (Hansen et al., 2013); 2000–2024
Wildfire occurrence	Annual burned-area indicator; captures fire disturbance and ecosystem stress.	MODIS MCD64A1 (Giglio et al., 2021); annual 2000–2024
Elevation and slope	Topographic characteristics influencing accessibility, agricultural suitability, and remoteness.	CGIAR-CSI SRTM V4 (Jarvis et al., 2008); static
Climate variables	19 bioclimatic variables (temperature, precipitation, seasonality, extremes) characterising long-run environmental conditions.	WorldClim V1 (Hijmans et al., 2005); static (1950–2000 climatology)

Variable / Group	Description	Source & Coverage
Global Safety Net (GSN) layers	5 selected conservation-priority indicators: (1) high-biodiversity areas, (2) intact wilderness areas, (3) climate-stabilisation areas, (4) wildlife corridors, and (5) WWF terrestrial ecoregions.	Global Safety Net (Dinerstein et al., 2020); static (2020)
Land cover	Annual dominant land-cover class (IGBP scheme); distinguishes forest, cropland, urban, wetland, and other types.	MODIS MCD12Q1 (Friedl and Sulla-Menashe, 2022); annual 2001–2024
Human Pressure, Economic Activity, and Infrastructure		
Population density	Gridded population counts; proxy for development intensity and opportunity cost of protection.	GPW v4 (Center for International Earth Science Information Network (CIESIN), Columbia University, 2018); 5-year intervals 2000–2020
Night-time lights (HNTL)	Harmonised night-time light intensity (DMSP/VIIRS); proxy for economic activity and built infrastructure.	HNTL (Li et al., 2020); annual 2000–2024
Road infrastructure	Major road network density; captures accessibility and development pressure.	GRIP (Meijer et al., 2018); static (2018)
Energy and extractive infrastructure	Power plants and oil & gas well/pipeline locations as proxies for industrial capital investment and development pressure.	Global Power Plant Database (Global Energy Observatory et al., 2019); Oil & Gas Infrastructure (Omara et al., 2023); static

Variable / Group	Description	Source & Coverage
Indigenous territories	Distance to boundaries of formally recognised indigenous and community lands; captures institutional overlap between conservation and indigenous rights.	LandMark (LandMark – Global Platform of Indigenous and Community Lands, 2023); static
Agricultural land value	Country-level FAO gross production value spatially allocated to agricultural pixels (cropland and pasture) using MODIS land-cover weights; proxies the economic opportunity cost of protecting productive land.	FAO FAOSTAT (Food and Agriculture Organization of the United Nations, 2024); annual 2000–2024
Governance and Institutional Context		
Government effectiveness & rule of law	WGI Government Effectiveness and Rule of Law scores; country-level measures of institutional capacity to translate policy commitments into on-the-ground designations.	World Bank WGI (World Bank, 2025); annual 2000–2024
Democratic governance	V-Dem Liberal Democracy Index; captures the quality of democratic institutions and civil society, which may shape conservation policy responsiveness.	V-Dem v15 (V-Dem Institute, 2025); annual 2000–2024
Executive political ideology	DPI executive ideology score (left/centre/right); controls for political-economy variation in conservation priorities across governments.	DPI 2020 (Scartascini et al., 2021); annual 2000–2020

Existing conservation context. Distance to the nearest PA boundary (`dist_wdpa`) captures spatial path dependence: designations cluster near the existing network as buffer zones and corridor links. Lagging by one year prevents the model from exploiting same-year PA information.

Environmental conditions. NDVI, deforestation, wildfire, elevation, slope, landcover, and WorldClim bioclimatic variables characterise biophysical suitability and ecological value. The GSN layers add globally standardised conservation-priority scores covering biodiversity hotspots, intact wilderness, and climate-stabilisation areas.

Human pressure and economic activity. Population density, night-time lights, and distances to roads, power plants, and oil and gas installations characterise the intensity of human land use and the opportunity cost of designation. Agricultural land value provides a direct economic proxy: high-value agricultural land is more costly to designate and thus expected to face stronger political resistance to protection. The variable is constructed by allocating each country’s annual gross agricultural production value (FAO FAOSTAT, in constant USD) across its crop and pasture pixels in proportion to their MODIS land-cover area, with cropland pixels weighted three times higher than pasture pixels to reflect the systematic land-value differential between these two agricultural types: US national average cropland values are approximately three times those of pasture (USDA NASS, 2025), a ratio broadly consistent with Brazilian evidence where cropland prices are roughly twice those of pasture (Richards et al., 2014). This allocates a country-level average value-per-km² to each agricultural pixel, capturing year-to-year changes in commodity prices and production patterns without requiring parcel-level data that is unavailable at global scale. The allocation assumes that gross value per unit area is uniform within land-cover class within a country, an approximation that understates variation across farm types but preserves the cross-country and cross-crop-type signal most relevant for transition-risk comparisons.

Institutional predictors. Distance to formally recognised indigenous and community lands (LandMark) captures the spatial overlap between indigenous territories and conservation expansion, a substantively important pattern in South America and Southeast Asia.

Country-level governance indicators (WGI Government Effectiveness, WGI Rule of Law, V-Dem Liberal Democracy Index, and DPI executive political ideology) capture the institutional capacity and political economy of each country. Cross-country variation in designation pace is at least partly attributable to institutional differences: stronger rule of law and greater government effectiveness are associated with the ability to convert international conservation commitments into on-the-ground designations (Watson et al., 2014).

Static versus dynamic predictors. Variables that are time-invariant, such as elevation, slope, WorldClim climatology, road networks, GSN layers, oil and gas infrastructure, and distance to indigenous territories, are treated as static and repeated across all years for a given cell. Time-varying variables, such as NDVI, deforestation, wildfire, population density, night-time lights, distance to PAs, agricultural land value (annual FAO production), and governance indicators (WGI, V-Dem, DPI; assigned by country-year) are updated annually. Treating some dynamic processes (road expansion, energy infrastructure growth) as static is a recognised limitation, discussed further in Section 6.

Missing values. Missing values arise from genuine data gaps (e.g. incomplete satellite coverage in cloud-persistent regions). Rather than imputing missing values, they are retained as NA sentinels and handled natively by the tree-based models, which route missing observations to a dedicated split branch during training and inference without distorting the observed feature distribution. For population density (available at five-year intervals) and night-time lights (a small end-of-series gap), forward-filling is applied: for any year t without a direct observation, the most recent prior-year value is carried forward within each cell. This preserves within-cell trends while avoiding interpolation artefacts between census rounds.

3.3 Harmonisation and Panel Construction

Reference grid and coordinate system. All analyses are conducted on a regular 1×1 km reference grid in the coordinate reference system EPSG:3857 (WGS 84 Web Mercator). The 1 km resolution is justified on three grounds. First, data compatibility: the key predic-

tors, including WDPa boundaries, WorldClim climate normals, GPW population density, and MODIS land cover, are all natively available at 1 km or coarser, so a finer grid would require interpolation without adding information. Second, model–data alignment: protected areas are policy decisions that operate at landscape scale, not sub-kilometre parcels; a 1 km cell is both the finest grain at which all relevant drivers can be jointly observed and a practical planning unit for area-based conservation targets. Third, consistency with the broader PA prediction literature, where 1 km is the established reference resolution (Greenhill et al., 2024; Mouillot et al., 2024). While EPSG:3857 introduces Mercator areal distortions at high latitudes (at $\varphi \approx 55^\circ\text{S}$ the scale factor is $k \approx 1.74$); these distortions apply uniformly to all input layers, so they cancel in any per-cell comparison and leave risk rankings unaffected.

A fixed land-only backbone raster defines the universe of cells eligible for analysis. Ocean pixels and pixels outside each continental study boundary are excluded uniformly from all datasets, ensuring that the risk set contains only land pixels with complete covariate coverage.

Reprojection and resampling. Every input raster is reprojected to the common 1 km backbone grid and temporally aligned to calendar years. For continuous variables (elevation, NDVI, population density, climate, night-time lights), bilinear interpolation is used during reprojection: it averages neighbouring pixels and produces smooth transitions, avoiding the block artefacts of simpler methods. For categorical and binary layers (land-cover class, PA status, conservation priority indicators), nearest-neighbour interpolation preserves discrete class boundaries. Satellite image collections with multiple acquisitions per year are collapsed to a single annual summary: annual median for continuous variables (reducing cloud contamination and sensor noise) and cumulative sum for fire events (accumulating burned area over the year).

Panel construction. Following spatial and temporal harmonisation, all layers are merged into a unified pixel–year panel. Each row corresponds to a single grid cell i in a specific year t and contains: (i) the binary target variable $\text{transition}_{i,t}$; (ii) the full set of static and dynamic predictor variables; and (iii) cell and year identifiers. The resulting South America panel contains approximately 413 million pixel–year observations across 20.8 million unique grid

cells observed over 25 annual periods (2000–2024). Comparable panels are constructed for the United States and Southeast Asia.

4 Methods

Figure 1 gives an overview of the end-to-end pipeline. The subsections below describe each stage in detail.

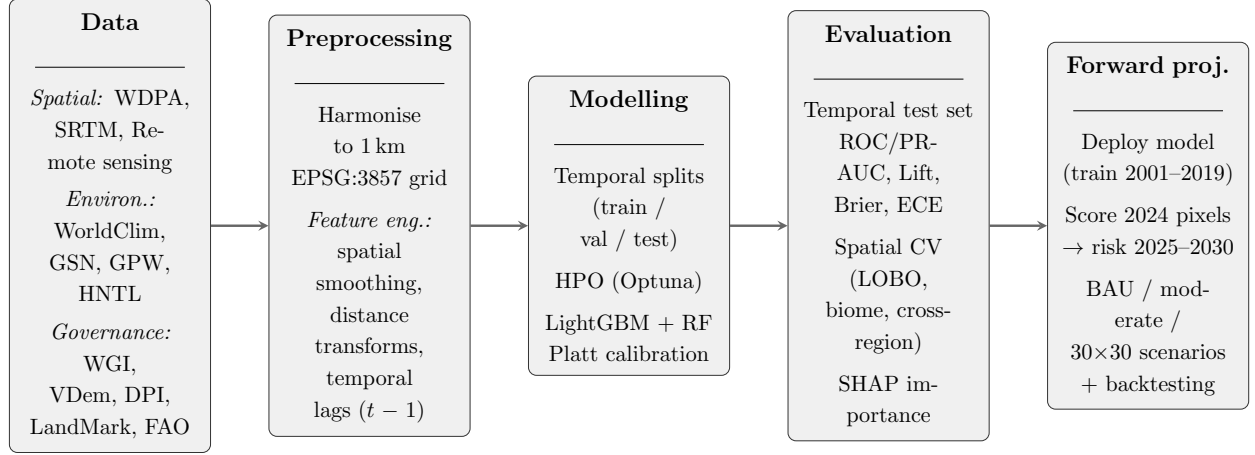


Figure 1: End-to-end pipeline overview.

4.1 Modelling Framework

Problem formulation. The objective is to estimate, for each currently unprotected 1 km² grid cell i observed in year t , the probability that cell i becomes newly designated as a protected area within the subsequent five years. Let $y_{i,t} \in \{0, 1\}$ denote whether cell i is protected in year t . The target variable is a five-year forward-looking binary indicator:

$$\text{transition}_{i,t} = \begin{cases} 1 & \text{if } y_{i,t} = 0 \text{ and } y_{i,t+k} = 1 \text{ for any } k \in \{1, \dots, 5\}, \\ 0 & \text{otherwise.} \end{cases} \quad (1)$$

A five-year horizon is chosen because it corresponds to the medium-term planning cycle of most conservation programmes and aligns with the 2025–2030 window of the 30×30 target.

Risk set and right-censoring. Only unprotected pixels ($y_{i,t} = 0$) are eligible for the transition and therefore included in the analysis. Once a pixel becomes protected, it exits the risk set permanently and does not contribute further observations. This prevents the model from learning to predict non-events for land that is already protected.

Right-censoring is a technical constraint arising from the finite data horizon. To observe the full five-year outcome window, the last year for which a label can be constructed is 2019 (using PA designations through 2024). All pixel-year observations beyond 2019 are excluded from model training and evaluation; the target variable is undefined for them because we cannot verify whether a transition occurs in the full subsequent five-year window. This cutoff year is denoted `LAST_LABEL_YEAR = 2019`.

Rare-event structure. Transitions into protected status are rare: fewer than 0.5% of eligible pixel-year observations record a positive outcome. This extreme class imbalance is not an artefact of data collection but reflects the true pace of PA expansion relative to the vast unprotected land area. Standard classifiers trained on such data tend to predict the majority class (non-transition) for almost all observations and achieve high nominal accuracy while being useless for identifying future designations (Davis and Goadrich, 2006). The modelling choices described in Sections 4.3–4.5 are designed specifically to address this challenge.

Relation to discrete-time hazard models. The panel formulation of Equation (1) is conceptually related to discrete-time hazard (or duration) models used in econometrics to study the timing of events such as unemployment spells, technology adoption, or firm bankruptcy (Allison, 1982; Singer and Willett, 1993). In the hazard interpretation, each unprotected pixel is “at risk” of designation in each year, and the model estimates the conditional probability of transitioning given that no transition has occurred yet. Unlike parametric hazard models, the tree-based machine-learning approach used here imposes no functional form on the relationship between predictors and the transition probability, and handles complex interactions among the 73 engineered covariates without prior specification of functional forms.

Prediction, not prescription. The framework learns *where governments have historically designated* PAs and extrapolates this pattern forward. It does not predict where PAs *should* be designated for maximum conservation impact. The two questions have different answers because real designation is not only shaped by biodiversity optimisation, but also by political feasibility and opportunity cost (Schuster et al., 2023). The gap between predicted and optimal designations is itself a substantive finding discussed in Section 6.

4.2 Feature Engineering

Raw cell-level features describe conditions at a single pixel. Because PA designation is influenced by conditions in the surrounding landscape, additional features are engineered to encode spatial context at multiple scales.

Multi-scale spatial smoothing. For a selected subset of time-varying predictors (NDVI, population density, night-time lights, deforestation intensity, and wildfire), uniform averaging kernels of size 4×4 , 16×16 , and 64×64 grid cells are applied, corresponding to neighbourhood extents of 4 km, 16 km, and 64 km respectively. These three scales are designed to capture: (i) immediate local context (4 km: adjacent land use, buffer zones); (ii) sub-regional landscape dynamics (16 km: agricultural frontier gradients, infrastructure corridors); and (iii) landscape-scale patterns (64 km: macro-regional biome boundaries, large-scale deforestation fronts). Including all three scales allows the model to select the most informative spatial context for each combination of predictor and location without imposing a single fixed neighbourhood size (Collart et al., 2025; McGarigal et al., 2016).

Distance-based features. In addition to smoothed averages, Euclidean distances to the nearest PA boundary, road segment, power plant, and oil and gas installation are included as features. These distance measures directly encode proximity to conservation and development infrastructure, capturing processes such as incremental PA network expansion and the deterrent effect of industrial activity on designation.

Temporal lag. All time-varying features, including the distance to existing PAs, are lagged by one year: features at year t reflect conditions recorded through end of year $t - 1$. This ensures that the model cannot use information about PAs designated in year t itself to predict the transition target for year t , which would constitute temporal leakage. For example, distance to the nearest PA at year t is computed using only the PA network as it stood at end of year $t - 1$, so designation events in year t do not inform the features used to predict those same events.

4.3 Machine Learning Models

Three modelling approaches are estimated for each region, in increasing order of complexity.

Static Suitability Baseline. A static suitability model is estimated on cross-sectional data from the year 2000: it learns to distinguish pixels that are already protected from those that are not, using only time-invariant geographic characteristics (elevation, biodiversity, climate, remoteness). Scores for the unprotected year-2000 pixels are then compared against PA designations observed between 2000 and 2024. Because this model ignores all temporal dynamics and uses a different target (current protection status rather than future transition), it establishes a meaningful lower bound: any gain over the static baseline quantifies the added value of temporal variation in covariates and the transition-specific training setup.

Random Forest. A Random Forest (Breiman, 2001) classifier is trained on the pixel-year panel. The algorithm builds an ensemble of 300 decision trees, each fitted to an independent bootstrap sample of the data, considering a random subset of features at every split. Predictions are obtained by averaging probabilities across all trees. This ensemble structure produces probability estimates that are more stable than any single tree and naturally provides feature importance measures from the average reduction in node impurity across splits. Class imbalance is handled via `class_weight="balanced_subsample"`, which reweights minority-class (transition) observations within each tree in inverse proportion to their frequency.

LightGBM. Light Gradient-Boosting Machine (LightGBM; Ke et al., 2017) is the primary reporting model throughout the analysis. Both LightGBM and Random Forest are fully trained and evaluated production models; results are reported for LightGBM by default because it attains superior performance on the calibration-sensitive metrics most relevant for transition-risk applications (PR-AUC and Brier score), while RF results serve as confirmation. Section 5.2.1 elaborates on this choice.

Gradient boosting builds an ensemble of decision trees sequentially: each tree is fitted to the negative gradient (pseudo-residuals) of the ensemble’s current loss, so successive trees progressively correct prediction errors. LightGBM introduces Gradient-based One-Side Sampling (GOSS), which retains all large-gradient observations and subsamples low-gradient ones, and Exclusive Feature Bundling (EFB), which groups mutually exclusive sparse features. These optimisations make it substantially faster than standard gradient boosting on large datasets without loss of accuracy, a critical advantage for panels of 100–350 million observations. Class imbalance is handled via `scale_pos_weight`, set equal to the ratio of negative to positive observations in the training set:

$$\text{scale_pos_weight} = \frac{N_{\text{neg}}}{N_{\text{pos}}}. \quad (2)$$

This upweights each minority-class (transition) observation in the binary cross-entropy loss, steering the boosting iterations toward correct identification of rare designations. The parameter is recomputed from the training data for each model run and is not tuned as a hyperparameter.

Hyperparameters for both models are selected via Bayesian optimisation using Optuna (Akiba et al., 2019) with a TPE sampler and median pruner, running up to 100 trials per region. The optimisation objective is PR-AUC on the held-out early-stopping split of the innermost cross-validation fold (see Section 4.5). Table 2 reports the key values for both models.

Table 2: Key hyperparameter values for the Random Forest (RF) and LightGBM (LGBM) transition models.

Parameter	RF	LightGBM	Notes
Number of trees / rounds	300	Tuned per region	Fixed from CV
Max depth	Unlimited	Tuned	LGBM: -1 = no limit
Min samples per leaf	1	Tuned	LGBM: <code>min_child_samples</code>
Class imbalance	<code>balanced_subsample</code>	<code>scale_pos_weight</code>	See Eq. (2)
Random seed	42	42	Fixed for reproducibility

4.4 Probability Calibration

Raw model outputs are scores, not calibrated probabilities. A raw score of 0.7 does not mean the pixel has a 70% chance of being designated; it only indicates that the pixel ranks above most others in the predicted risk distribution. This distinction is consequential for transition-risk applications: portfolio screening or regulatory stress-testing requires not just a ranking of risky assets but also an estimate of *how risky*, expressed as a probability that admits direct comparison across assets and thresholds.

The gap between raw scores and true probabilities arises because the class-weighting scheme (Equation 2) reshapes the effective class distribution seen by the model. After reweighting, the model optimises against a balanced distribution and learns to assign higher scores than the true probability of the minority class warrants.

To correct this, calibration is performed using Platt scaling (Platt, 1999; Niculescu-Mizil and Caruana, 2005): a logistic regression is fitted with the raw model scores as input and the true binary labels as the outcome, using out-of-fold cross-validation predictions so that no test-set information enters the calibration. This logistic function maps raw scores to calibrated probabilities whose mean matches the empirically observed transition rate in each score band. Isotonic regression (Zadrozny and Elkan, 2002) is computed as a robustness check. Calibrated probabilities are clipped to $(10^{-6}, 1 - 10^{-6})$ for numerical stability. All probability maps and risk scores reported in this thesis use Platt-calibrated predictions.

4.5 Training, Validation, and Test Design

Temporal splits. Model training and evaluation strictly preserve temporal order to ensure that all reported metrics reflect the model’s ability to forecast *future* designations using only *past* information. The training design consists of three stages:

Hyperparameter optimisation (Optuna HPO). Optuna (Akiba et al., 2019) runs up to 100 trials, each fitting LightGBM or Random Forest with a candidate hyperparameter configuration and evaluating PR-AUC on the held-out early-stopping split within three expanding-window temporal cross-validation folds:

- Fold 1: train 2001–2008, early-stop validation 2009–2011.
- Fold 2: train 2001–2011, early-stop validation 2012–2013.
- Fold 3: train 2001–2013, early-stop validation 2014–2016.

Year 2000 is excluded from all folds: the WDPA records it as a cross-sectional baseline in which all pre-existing PAs appear as newly designated, not as true annual transition events. The best hyperparameter set is selected based on average out-of-fold PR-AUC across the three folds.

Final evaluation model. Using the selected hyperparameters, a final LightGBM model is trained on all non-test-period data (2001–2016). During each CV fold, early stopping (patience 500 rounds) was used to find the number of boosting rounds that maximises PR-AUC on the fold’s early-stop split; the final model trains for a fixed number of rounds equal to the training-size-weighted average of these fold-level best iterations, scaled by 1.05 to account for the larger effective training set, and no lower than the best iteration of the longest fold (Fold 3). This procedure determines training depth without exposing the test period.

Test period. The test set covers 2017–2019, held out entirely throughout hyperparameter search and model fitting. The upper boundary 2019 coincides with `LAST_LABEL_YEAR`: all pixel–year observations from 2020 onwards are right-censored. Every metric reported in Section 5 is an out-of-sample estimate based on this held-out test period.

Leakage prevention. Multiple safeguards prevent future information from reaching the model. (i) All time-varying features are lagged by one year, so the covariate vector at year t reflects conditions recorded through end of year $t - 1$ (Section 4.2). (ii) Distance to existing PAs is computed using only the PA network as of year $t - 1$, so same-year designation events cannot leak into the distance feature for that year. (iii) Hyperparameters are tuned using the Optuna HPO procedure described above, whose CV folds do not touch the test period; the test set is never used to guide any modelling decision. (iv) Probability calibrators are fitted exclusively on out-of-fold predictions generated during the CV folds, not on test data.

Temporal sample weighting. Conservation-policy dynamics evolve over time, and recent designation patterns are more informative for forecasting the near future than patterns from two decades ago. Training observations are therefore assigned a linearly increasing sample weight: year 2001 receives weight 0.5 and the last training year receives weight 1.0, with all intermediate years interpolated proportionally. This places greater emphasis on recent patterns while retaining the full historical record.

Class imbalance handling. Transitions are rare ($\approx 0.3\text{--}0.5\%$ of eligible pixel-year observations). No oversampling or undersampling is applied; the empirical frequency is preserved. Instead, class weights (LightGBM: `scale_pos_weight`; Random Forest: `balanced_subsample`) adjust the contribution of minority-class observations to the loss function. This approach avoids distorting the spatial or temporal structure of the data.

Reproducibility. All random components (tree construction, bootstrap sampling, Optuna trial initialisation) are controlled via a fixed random seed (42 throughout). The full pipeline is implemented in Python 3.12 using scikit-learn (Pedregosa et al., 2011), LightGBM (Ke et al., 2017), and GeoPandas, with intermediate outputs stored in Apache Parquet format for efficient column-oriented access. Large-scale model training (involving panels of 100–350 million rows) is executed on the ETH Zurich Euler high-performance computing cluster. Jobs are submitted as SLURM array tasks, which schedule independent parallel workloads (e.g. one array task per LOBO biome fold or per forward-prediction scenario) across the cluster’s compute nodes, reducing wall-clock time from days to hours.

4.6 Evaluation Metrics

Because the positive class constitutes fewer than 0.5% of observations, accuracy-based metrics (e.g. overall accuracy, F_1 -score) are uninformative: a model that predicts zero transitions everywhere achieves $> 99.5\%$ accuracy (Davis and Goadrich, 2006). Evaluation therefore focuses on metrics that are robust to severe class imbalance and that directly reflect the ranking quality of predicted risks.

Precision–Recall AUC (PR-AUC). The primary metric. The precision–recall curve plots precision (fraction of top-ranked pixels that actually transition) against recall (fraction of all transitions captured) as the classification threshold is varied:

$$\text{Precision}(k) = \frac{TP(k)}{TP(k) + FP(k)}, \quad \text{Recall}(k) = \frac{TP(k)}{TP(k) + FN(k)}. \quad (3)$$

Here $TP(k)$, $FP(k)$, and $FN(k)$ denote true positives, false positives, and false negatives at threshold k . PR-AUC is the area under this curve, integrated across all thresholds. Under severe class imbalance, PR-AUC is a more informative summary of discriminative performance than ROC-AUC because it places weight on the model’s ability to correctly rank positive observations at the top of the score distribution (Davis and Goadrich, 2006).

ROC-AUC. ROC-AUC measures the probability that a randomly chosen positive observation receives a higher predicted score than a randomly chosen negative. A value of 0.5 corresponds to random ranking and 1.0 to perfect discrimination. Although ROC-AUC is less sensitive than PR-AUC under extreme class imbalance, since it is dominated by the large number of true negatives; it provides a useful reference point for comparing model quality across settings with different base rates and for detecting degenerate or anti-predictive models.

Precision at top- $k\%$ and lift. For policy use, the most relevant question is: “if I inspect the top $k\%$ of predicted-risk pixels, what fraction will actually be designated?” This is precision at top $k\%$ of the score distribution ($P@k\%$). To account for differences in base

rates across evaluation splits, we also report *lift*:

$$\text{Lift@}k\% = \frac{P@k\%}{\text{baseline positive rate}}. \quad (4)$$

A lift of 60 at the top 1% means that screening the top 1% of pixels yields 60 times more future designations per unit area than random selection. Lift is reported at $k \in \{1, 5, 10\}$.

Brier score. The mean squared error of calibrated predicted probabilities:

$$\text{Brier} = \frac{1}{N} \sum_{i,t} (\hat{p}_{i,t} - y_{i,t})^2. \quad (5)$$

A lower Brier score indicates better-calibrated and more accurate probability estimates. The Brier score is bounded between 0 (perfect) and 1 (maximally wrong). Under a positive rate of π , a naive model that predicts the base rate $\hat{p} = \pi$ everywhere achieves a Brier score of $\pi(1 - \pi)$, which for $\pi \approx 0.008$ equals approximately 0.0079. A well-fitted model must beat this threshold; the Brier score of the calibrated LightGBM model (0.00421) represents a nearly 50% reduction relative to this no-information baseline.

Expected Calibration Error (ECE). ECE measures the average discrepancy between predicted probabilities and observed frequencies, computed by binning predictions into equal-width intervals and comparing the average predicted probability to the observed transition rate within each bin:

$$\text{ECE} = \sum_{b=1}^B \frac{|n_b|}{N} |\bar{p}_b - \bar{y}_b|, \quad (6)$$

where n_b is the number of observations in bin b , $\bar{p}_b$ is their mean predicted probability, and $\bar{y}_b$ is their observed transition rate. ECE is computed both before and after Platt scaling to directly quantify the calibration improvement achieved by post-hoc recalibration (Niculescu-Mizil and Caruana, 2005). A lower ECE indicates that predicted probabilities more faithfully track observed transition rates across the full range of predicted scores.

4.7 Feature Importance (SHAP)

To explain which features drive individual predictions, SHapley Additive exPlanations (SHAP; Lundberg and Lee, 2017) are computed for the LightGBM model using the tree-explainer algorithm. SHAP values are grounded in cooperative game theory: for each observation, the SHAP value of a feature equals its average marginal contribution to the model’s output across all possible orderings in which features could be introduced. A positive SHAP value for a given pixel–feature pair indicates that the feature’s value at that pixel pushes the predicted probability above the mean; a negative value indicates suppression of risk. The mean absolute SHAP value across all test-set observations is used to rank features by their average contribution to model predictions. This analysis is computed on the temporal test set (2017–2019) using the calibrated LightGBM model, and is separate from the feature importance computed during Optuna HPO (which uses impurity-based importance from the CV folds). Using the held-out test set for SHAP analysis is appropriate because it is the same data used to compute all reported benchmark metrics, so the feature attributions reflect model behaviour under genuine out-of-sample conditions without introducing any additional leakage beyond what is already present in the evaluation design.

4.8 Spatial Robustness Evaluation

In addition to temporal out-of-sample evaluation, spatial robustness is assessed through a three-layer framework that tests whether the model’s predictive patterns hold across geographic settings it was not explicitly trained on (Ploton et al., 2020).

4.8.1 Leave-One-Biome-Out (LOBO)

LOBO cross-validation, adapted from spatial leave-one-group-out procedures (Ploton et al., 2020; Roberts et al., 2017), provides the strictest within-continent generalisation test. Biomes are defined using the WWF terrestrial ecoregion classification from the Global Safety Net (GSN) layer. For each biome b , a separate model is trained on all observations from all *other* biomes and evaluated on the held-out biome’s test-period pixels. A biome with low LOBO performance indicates that its designation patterns are structurally different from those in

the rest of the continent; the model learned patterns that do not transfer to that biome’s ecological and governance context.

ROC-AUC is reported as the LOBO metric because it directly measures cross-biome *ranking* quality without being sensitive to the specific base rate of the held-out biome. Under LOBO, each biome has a different positive rate, making PR-AUC values across biomes incomparable. ROC-AUC, by contrast, is base-rate invariant and provides a consistent measure of whether the model’s score ordering is informative in biomes it has never seen.

4.8.2 Biome-Stratified Evaluation

The standard (full-data) production model is scored on the test period, and its performance metrics are broken down by biome. Unlike LOBO, this diagnostic does not test generalisation to unseen biomes: the model was trained on all biomes simultaneously. Instead, it reveals where within a continent the model performs well or poorly, specifically which biomes produce the clearest predictive signal versus where prediction is harder even when the biome’s own history was available for training. This biome-level performance map is useful for communicating the reliability of risk scores in specific geographic areas.

4.8.3 Cross-Continental Transfer

Cross-continental transfer provides the most demanding spatial test: a model trained on one continent is scored on another without any retraining. Transfer is performed in all six directions (SA→USA, USA→SA, SA→SEA, USA→SEA, SEA→SA, SEA→USA) using the common feature set available in all three regions. High transfer performance would imply that the model has learned universal ecological and governance drivers of PA designation. Near-random transfer performance implies that designation patterns are continent-specific and shaped by regional political economies, itself a substantive result about the limits of cross-context generalisation for conservation risk models.

4.9 Forward Prediction to 2030

Deployment model. The forward prediction setup is identical to the evaluation model in every respect, using the same hyperparameters, same features, same class-weighting scheme, and same Platt calibration procedure, with one difference: the training window is extended to include all non-censored years. A deployment model is therefore trained on 2001–2019 (all years for which the five-year transition target can be constructed without right-censoring), using the same hyperparameter configuration selected by the Optuna HPO stage and the same number of boosting rounds as the evaluation model. Because the deployment model trains on all labeled data including the 2017–2019 test period, no held-out set is available for calibration. An auxiliary model trained on 2001–2016 is therefore used to score 2017–2019 as a held-out proxy, and the resulting out-of-fold predictions are used to fit the Platt and isotonic calibrators. This is a practical approximation that introduces a small mismatch between the calibration context and the deployment context; the mismatch is disclosed in the backtesting section.

Forward inference. Inference is performed on the 2024 covariate snapshot for all pixels unprotected as of 2023. The deployment model assigns each candidate pixel a calibrated probability of becoming protected between 2025 and 2030. This constitutes a strict temporal out-of-sample forecast: no 2024 information enters model training. On temporal alignment: all time-varying features are lagged by one year (Section 4.2), so the 2024 covariate snapshot reflects conditions recorded through end-2023. Since unprotected status is evaluated at end-2023 (“unprotected as of 2023”), the effective reference date for the five-year lookahead is the beginning of 2025, placing the forecast horizon squarely within the 2025–2030 window of the 30×30 target.

Policy scenarios. Three scenarios are derived from the resulting probability surface by varying the cumulative area designated:

- **Business-as-usual (BAU):** cumulative new area equals the observed 2019–2024 designation volume, projecting the recent empirical designation rate forward.

- **Moderate:** cumulative area targets the midpoint between each region’s current coverage and 30%, applied proportionally.
- **30×30:** cumulative area targets full 30% terrestrial coverage by 2030.

Under each scenario, the model selects the highest-probability pixels to fill the required area, producing spatially explicit designation maps. Hotspot clusters are identified via spatial clustering (Ester et al., 1996) on the top-probability pixels.

Backtesting. Methodological validity is established by backtesting: pseudo-forecasts are generated at historical origins $T \in \{2009, 2011\}$, each time training on 2001–($T - 1$), scoring unprotected pixels in year T , and evaluating against realised designations in years $T + 1$ through $T + 5$. This walk-forward evaluation empirically verifies that the forward deployment procedure generalises to out-of-sample designation events.

5 Results

5.1 Descriptive Statistics and Transition Dynamics

The South America panel contains approximately 413 million pixel–year observations spanning 20.8 million unique 1 km² grid cells over 25 annual periods (2000–2024). By 2024, 25.6% of South America’s terrestrial area is formally protected, up from approximately 15% in 2000. As a consequence of this cumulative expansion, the risk set shrinks from roughly 20.8 million eligible pixels in 2000 to approximately 15.7 million by the end of the sample.

Transitions from unprotected to protected status are rare events. The average transition rate across the full South America panel is approximately 0.5–0.8% per pixel–year, implying that a given pixel has less than a 1-in-100 chance of being designated in any given year. This rate is not constant over time: designation rates were higher during 2000–2013, a period of active PA expansion in the Amazon and Cerrado, and declined thereafter. In the held-out test period (2017–2019), 380,064 transitions occur among approximately 47.7 million eligible pixel–year observations, corresponding to a positive rate of approximately 0.8%.

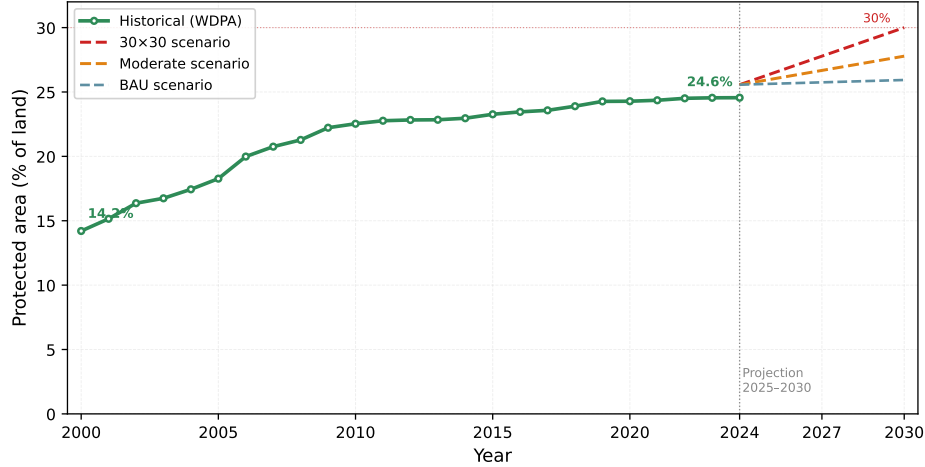


Figure 2: South America: cumulative protected-area coverage (2000–2024) with scenario projections to 2030. Historical data (circles) are from annual WDPA rasters. Dashed lines show three forward scenarios from the 2024 baseline (25.6%): business-as-usual (BAU, +0.061%/yr), moderate target (27.8% by 2030), and the 30×30 target (30% by 2030). Coverage grew rapidly during 2000–2013 and has plateaued well below the 30×30 threshold; reaching the target requires a 12× acceleration relative to the BAU trajectory.

The United States panel covers approximately 200 million pixel–year observations. Current protection stands at 7.7%, and the test-period transition rate is approximately 0.04% (an order of magnitude lower than South America), reflecting the US’s severely constrained designation pace relative to its 30% target. Southeast Asia covers approximately 100 million pixel–year observations, with current PA coverage of 14.0% and a test-period transition rate of approximately 0.6%.

These patterns reflect three defining features of the data: (i) a large risk set; (ii) strong temporal heterogeneity in transition rates across regions and time periods; and (iii) extreme class imbalance, most pronounced in the US and in later sample years.

5.2 South America: Model Performance

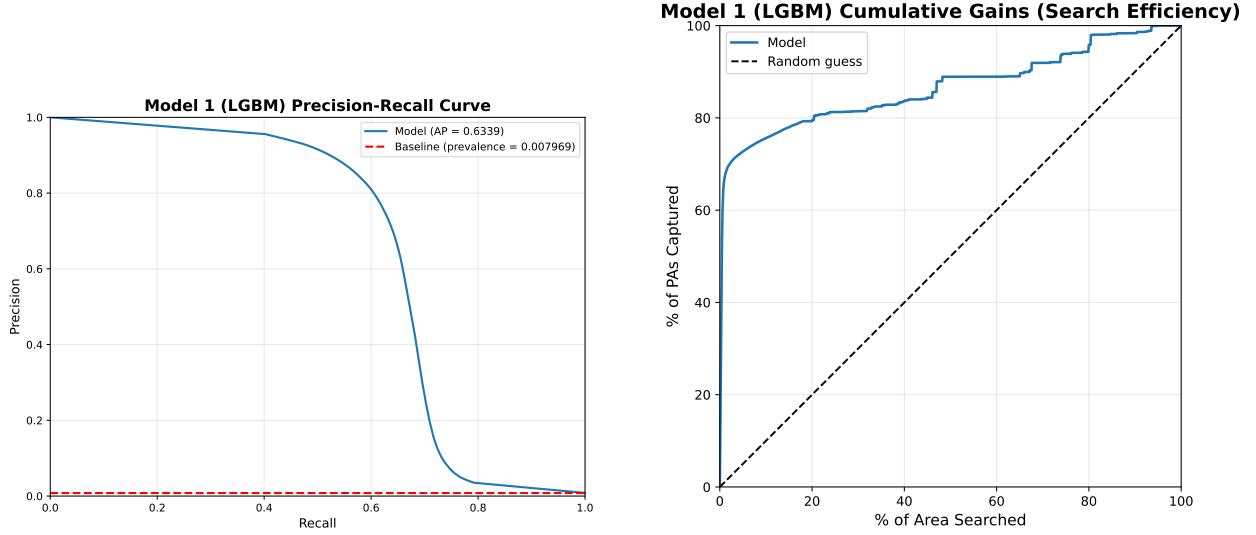
5.2.1 Discrimination Metrics

Table 3 reports temporal test-set performance for the LightGBM and Random Forest transition models applied to South America (test period: 2017–2019).

Table 3: Predictive performance on the temporal test set (2017–2019) for South America. Baseline positive rate: 0.797%.

Metric	LightGBM	Random Forest
ROC-AUC	0.897	0.927
PR-AUC	0.645	0.634
Precision @ top 1%	0.532	0.539
Lift @ top 1%	66.8×	67.6×
Precision @ top 5%	0.116	0.121
Lift @ top 5%	14.5×	15.2×
Brier score	0.00421	0.00458
ECE (post-calibration)	0.0081	0.0109

The LightGBM model achieves strong discrimination: a ROC-AUC of 0.897 indicates that a randomly drawn transition pixel receives a higher risk score than a randomly drawn non-transition pixel 90% of the time. The PR-AUC of 0.645 is the more informative metric under extreme imbalance: compared to a random classifier, which would achieve a PR-AUC equal to the base rate (0.008), the model represents an 80-fold improvement in precision-recall performance. At the most policy-relevant threshold, the top 1% of predicted risk pixels yield 66.8 times as many future designations per unit area as random selection (Lift@1% = 66.8×). The Brier score of 0.00421 measures mean squared probability error and is 47% below the no-information baseline of $\pi(1 - \pi) \approx 0.0079$, confirming that calibrated predictions carry genuine probability accuracy. The post-calibration ECE of 0.0081 indicates that predicted probabilities deviate from observed transition rates by an average of 0.81 percentage points across score bins, a low miscalibration for a rare-event classifier.



(a) Precision–Recall curve. The dashed line shows the random-classifier baseline ($\approx 0.8\%$ positive rate). The curve lies far above the baseline across the full threshold range.

(b) Cumulative gains chart. The curve shows the fraction of all test-period transitions captured as a function of the fraction of pixels screened, ranked by predicted probability.

Figure 3: Model discrimination for the South America LightGBM on the temporal test set (2017–2019). *Left:* Precision–Recall curve. *Right:* Cumulative gains chart — at the top 1% of pixels, the model captures $\approx 66.8\%$ of all transition events (Lift $66.8\times$).

Figure 3 shows the Precision–Recall curve (left panel) and cumulative gains chart (right panel) for the LightGBM model. As the classification threshold decreases from right to left, precision (fraction of flagged pixels that actually transition) falls while recall (fraction of all transitions flagged) rises. The curve lies far above the dashed baseline throughout, confirming that the model’s advantage over random selection is not an artefact of a single threshold but holds across the entire operating range. The cumulative gains chart provides a complementary view from the perspective of recall: at the top 1% of pixels, the model captures $\approx 66.8\%$ of all test-period transitions (Lift $66.8\times$).

Both LightGBM and Random Forest are fully trained and evaluated production models, and their results confirm each other: the two classifiers produce broadly similar performance across all metrics. The Random Forest achieves a higher ROC-AUC (0.927 vs. 0.897), while LightGBM leads on PR-AUC (0.645 vs. 0.634), Brier score (0.00421 vs. 0.00458), and ECE (0.0081 vs. 0.0109). For transition-risk applications, PR-AUC and probability calibration are the more actionable metrics, as they measure how well the model identifies and quantifies rare events at specific operating thresholds rather than across a majority-class-dominated

ranking. LightGBM is therefore reported as the primary model in the remainder of the analysis, with RF results provided for confirmation; all substantive conclusions are robust to this choice.

5.2.2 Event-Level Coverage

Pixel-year metrics such as PR-AUC penalise any error in timing: a model that correctly identifies the *location* of a future designation but places it one year off still receives no credit for that positive. For a policy application such as screening land for transition risk, the question of interest is spatial rather than temporal: does the model identify the right *places*, regardless of precisely when designation occurs within the next five years?

Event-level coverage answers this question directly: for each unique PA establishment event within the five-year prediction window, the metric checks whether that event’s location falls within the top-1% risk zone. The prediction window for each test-year observation extends five years forward: 2017 test rows look ahead to 2022, 2018 to 2023, and 2019 rows to 2024. This definition is identical to the `transition_01_win5` label used for all pixel-year metrics, making event-level recall directly comparable to Recall@K. Under this aggregation, the LightGBM model achieves a **Recall@1% of 77.4%** and a **Precision@1% of 79.8%**: nearly four in five of the top-1% predicted pixels actually became protected areas, and three in four of all new protected areas fell within just 1% of eligible land (approximately 159,000 km² out of 15.9 million km²). In other words, a risk analyst screening fewer than 1% of candidate parcels would flag roughly three-quarters of all future designation events well in advance, while maintaining a precision rate close to 80%. This combination of high precision *and* high recall at an extremely concentrated threshold is the defining result: the moderate pixel-year PR-AUC reflects the inherent difficulty of predicting the exact year of a politically-driven administrative decision, not a failure of spatial targeting.

5.2.3 Probability Calibration

Before Platt scaling, raw LightGBM scores are systematically overconfident in the high-probability range: the model assigns scores of 0.9 to observations that become protected only 60% of the time. After calibration, the ECE drops to 0.0081, confirming that calibrated

probabilities closely track observed transition rates across all bins of the score distribution. Reliability diagrams before and after calibration are provided in Appendix B.

The practical implication is that a pixel with a calibrated predicted probability of 0.03 has a 3% chance of being designated within five years, a statement that is meaningful for portfolio screening and risk quantification, not merely for ranking.

5.2.4 Feature Importance (SHAP)

Figure 4 shows the SHAP summary plot for the South America LightGBM model on the test set. Each row is one feature, sorted from top (highest mean absolute SHAP value) to bottom. For each observation, the horizontal position of a dot shows that feature’s SHAP value, measuring how much it pushed the prediction above (positive) or below (negative) the mean prediction. Dot colour encodes the raw feature value (red = high, blue = low). The plot simultaneously conveys both the average importance of a feature and the direction and nonlinearity of its effect.

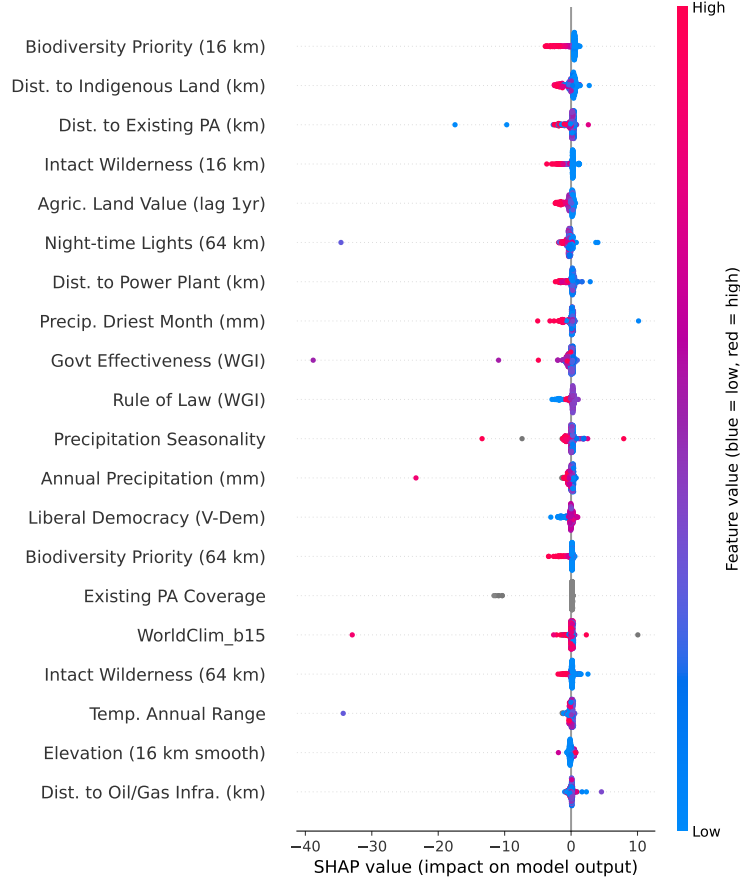


Figure 4: SHAP summary plot for the South America LightGBM model (test set 2017–2019). Features are sorted by mean absolute SHAP value (top = most important). Each dot represents one pixel–year observation; horizontal position is the SHAP value (positive = increases predicted designation probability; negative = decreases it); the colour bar on the right indicates the raw feature value (blue = low, red = high). Feature names have been translated from internal codes to interpretable labels.

The SHAP rankings partly confirm and partly qualify the simple opportunity-cost narrative. Three co-determining channels emerge:

Ecological conservation priority. The two leading individual features are High Biodiversity Areas at 16 km and 64 km smoothing (`GSN_b2_smooth16`, `GSN_b2_smooth64`) and Intact Wilderness at 16 km (`GSN_b3_smooth16`). Their positive SHAP values indicate that pixels embedded in globally significant biodiversity and wilderness areas attract designation, consistent with the ecological-value channel documented in conservation siting studies (Mouillot et al., 2024). The use of spatially smoothed versions of these layers, rather than the raw pixel-level binary indicator, confirms that designation is a landscape-scale process that responds to the *ecological neighbourhood*, not just the individual pixel.

Institutional capacity. WGI Rule of Law (`policy_b4`) is the second most important individual feature overall, comparable in magnitude to the top GSN layers, followed by V-Dem Liberal Democracy Index (`policy_b2`) and WGI Government Effectiveness (`policy_b3`). Their positive SHAP values indicate that designation is systematically more likely in countries with stronger institutions, independently of ecological conditions. This finding is consistent with the view that institutional capacity determines whether international conservation commitments are translated into on-the-ground designations (Watson et al., 2014). Because these are country-level indicators assigned to every pixel within a country, they capture a governance-level component of designation risk that cannot be explained by local ecological or economic characteristics alone.

Economic opportunity cost and path dependence. Agricultural land value (`economic_value_lag`) is a mid-tier predictor with predominantly negative SHAP values: high-value agricultural pixels are systematically suppressed from high-risk zones, confirming the opportunity-cost mechanism of Joppa and Pfaff (2009). Distance to indigenous territories (`dist_indigenous`) has mixed-sign SHAP values, reflecting the dual role of indigenous lands: they can both attract neighbouring designations (conservation buffer zones) and absorb designation pressure directly through community-based protections. Distance to the existing PA network (`dist_wdpa`) contributes positively at intermediate distances, capturing the incremental expansion of PA buffers. Infrastructure variables (night-time lights, roads, population density) appear at a lower tier but with consistent negative SHAP values, confirming that economic activity deters designation.

The overall picture is one of *complementarity*: designation is most likely where ecological conservation value is high, institutional capacity is present, and economic opportunity cost is low. These three conditions do not generally co-occur randomly: remote, biodiverse areas tend to have lower agricultural value, which is precisely why the literature characterises historical PA siting as biased toward economically marginal land (Joppa and Pfaff, 2009; Venter et al., 2018). The SHAP evidence confirms this mechanism while adding that institutional quality is an equally important and separately identifiable driver. SHAP dependence plots for the four most important features are provided in Appendix F.

5.2.5 Spatial Risk Maps

Aggregate metrics confirm that the model discriminates well on average, but the primary deliverable is a spatially explicit risk surface. Figure 5 maps the top 1% of predicted pixels across South America, overlaid with observed PA designations within the five-year prediction window (2017–2024). The observed category spans the full horizon that the model is trained to predict: all designations that occurred within five years of any test-period observation year (2017, 2018, or 2019), covering 2017–2022, 2017–2023, and 2017–2024 respectively. This means the map’s recall is directly comparable to the Recall@1% metric reported in the benchmark table; both denominators are identical. Of the total PA establishments captured, a subset occurred during the immediate test years (2017–2019) and the remainder in the subsequent window (2020–2024), demonstrating that the model’s risk signal persists beyond the held-out evaluation period.

High-risk pixels form spatially coherent clusters rather than a diffuse pattern, concentrated in the Amazon basin periphery, the Cerrado–Amazon transition zone, the Andean foothills, and parts of Chile’s southern regions. At Precision@1% = 79.8%, almost every cluster on the map corresponds to a place where a PA was in fact established; the model is not simply flagging the most biodiverse or remote areas in general, but identifying the specific locations where governments subsequently acted. The strong spatial correspondence between predicted high-risk zones and realised designations confirms that the model produces geographically meaningful risk concentrations, not merely a good score on abstract metrics.

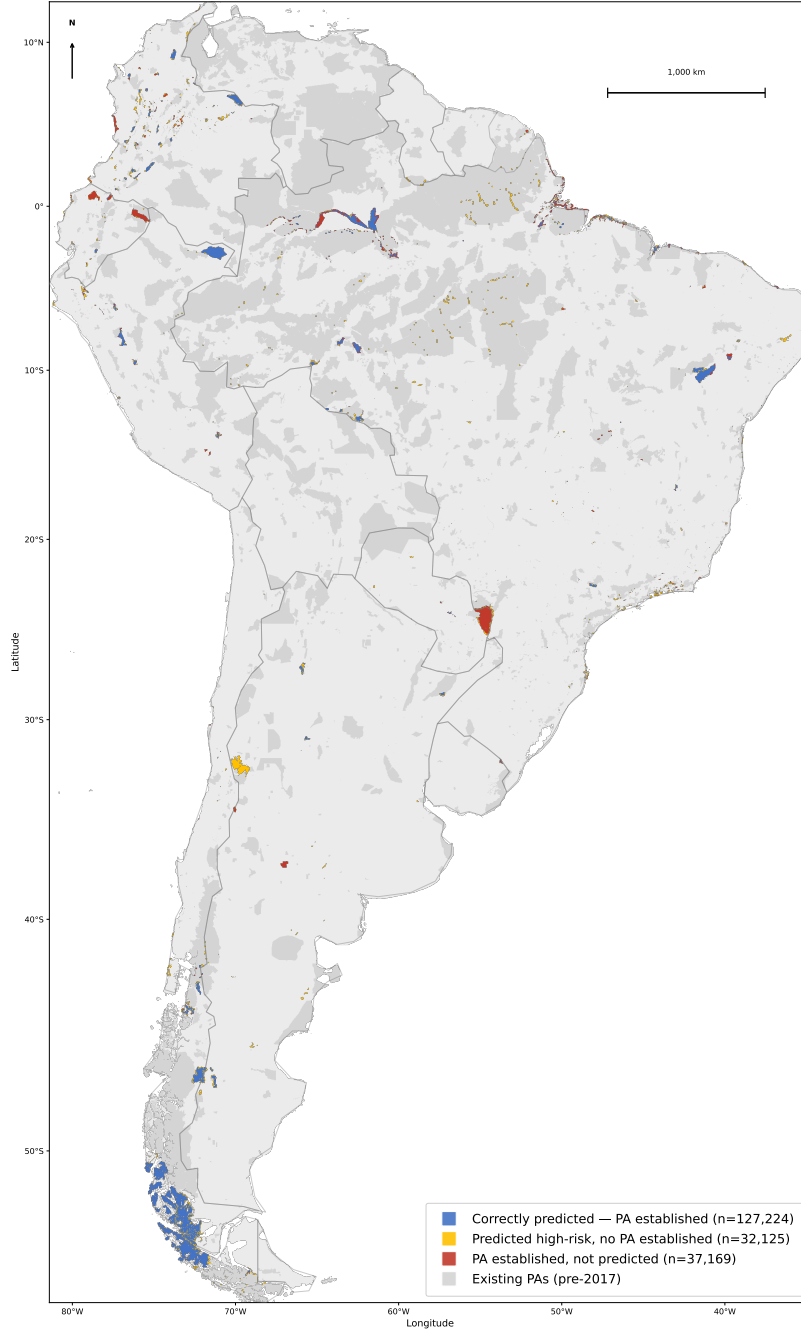


Figure 5: South America LightGBM risk map: top 1% of predicted pixels ($\text{Recall@1\%} = 77.4\%$, $\text{Precision@1\%} = 79.8\%$). **Blue:** correctly predicted, with the pixel in the top-1% risk zone *and* PA established within the five-year prediction window (true positive). **Gold:** predicted high-risk but no PA established within the window (unconfirmed prediction). **Red:** PA established within the window but pixel not in the top-1% zone (model miss). **Grey:** pixels already protected before the test period (excluded from the risk set). The observed category covers all PA designations within the five-year prediction horizon (up to 2024 for 2019 test rows), consistent with the `transition_01_win5` label used for all benchmark metrics. Pixel markers are slightly enlarged for print legibility.

The continuous probability map (Figure 6) shows a strongly right-skewed distribution:

the vast majority of pixels receive near-zero calibrated probability, while a small upper tail concentrates substantially elevated risk. The spatial autocorrelation of high-risk zones confirms that the model captures landscape-scale processes rather than pixel-level noise.

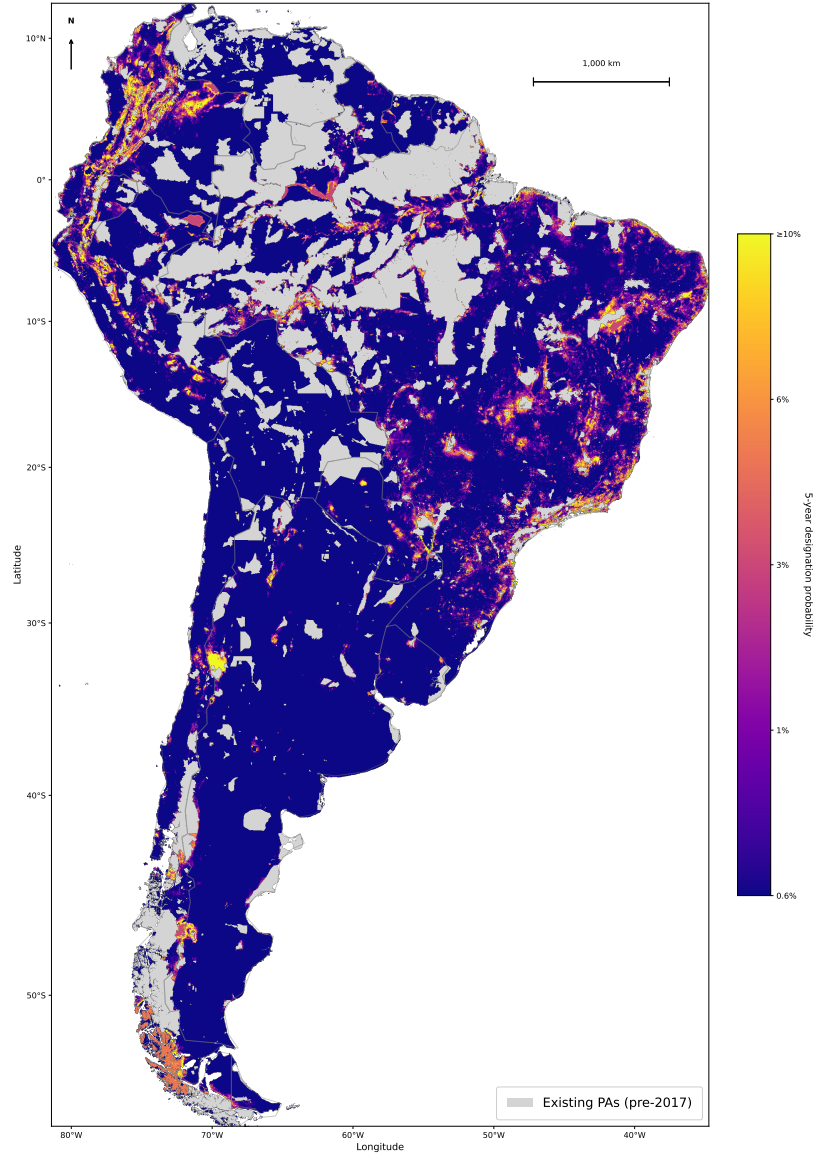


Figure 6: Continuous calibrated transition probabilities from the South America LightGBM model. Each pixel is coloured by its predicted five-year designation probability (square-root stretch from the 25th to the 98th percentile; dark purple = low probability, yellow = high probability). Grey pixels are existing protected areas excluded from the risk set. Probabilities reflect covariate values at the end of the test period.

5.2.6 Spatial Generalisation (LOBO)

The LOBO evaluation asks whether the patterns learned on other biomes generalise to a held-out one (see Section 4.8.1 for the methodological rationale). PR-AUC is used as the evaluation metric, consistent with the main model evaluation, because it is more informative under the extreme class imbalance typical of held-out biomes. The mean LOBO PR-AUC across nine South American biomes is 0.216, with substantial variation across biome types.

Performance varies markedly across biomes (Figure 7). Deserts and Xeric Shrublands (principally the Atacama) achieve a LOBO PR-AUC of 0.979: this biome’s high positive rate (10.4%, driven by the concentration of Chilean national parks) and distinctive combination of aridity, elevation, and low population density provide a strong-enough signature for models trained on other biomes to generalise. By contrast, Temperate Broadleaf and Mixed Forests (Patagonian Chile) achieves a near-random LOBO PR-AUC of 0.035 against a positive rate baseline of 2.0%. This is consistent with institutional drivers in Patagonia not represented in the feature set: much of the Chilean national park network in this biome was established through presidential decree and private conservation transactions (e.g. the Tompkins Conservation land donations), which do not follow the accessibility and land-value gradients that dominate designation elsewhere.

These biome-level contrasts are informative rather than simply disappointing: they identify regions where standard geospatial predictors are insufficient and where institutional data (land tenure, political mandates, NGO activity) would be needed to improve predictability.

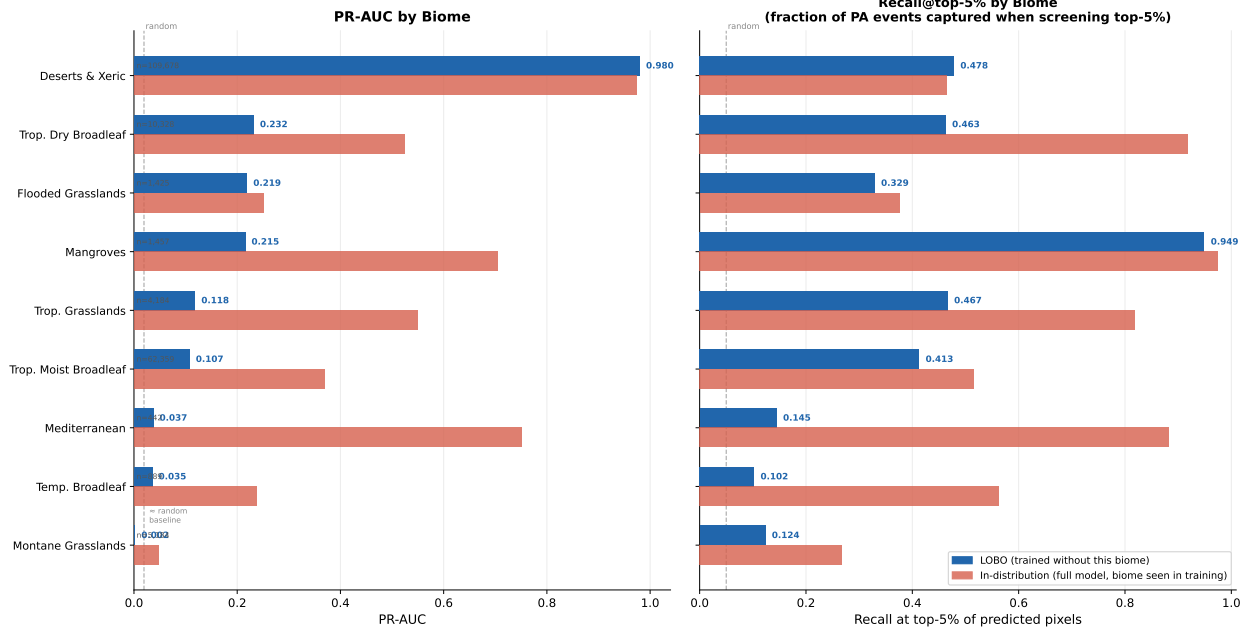


Figure 7: Per-biome LOBO PR-AUC for the South America LightGBM model (test set 2017–2019). The mean LOBO PR-AUC across nine evaluable biomes is 0.216. The dashed vertical line marks the approximate random-classifier baseline (mean biome positive rate); the Deserts & Xeric Shrublands biome stands out with PR-AUC 0.979 owing to its distinctively high positive rate (10.4%) and strong ecological signature.

5.3 United States: Robustness Check

The US results confirm that the predictive framework extends to a structurally different institutional context. The overall LightGBM ROC-AUC on the US test set is 0.947, indicating strong overall rank discrimination. However, the PR-AUC is only 0.093, reflecting the extreme class imbalance in the US test set ($\approx 0.04\%$ positive rate, 20 times rarer than South America). The high ROC-AUC alongside the very low PR-AUC reflects the known divergence of these two metrics under extreme imbalance (Davis and Goadrich, 2006): the model ranks positives above negatives in aggregate but struggles to maintain high precision at any fixed threshold. The Lift at top 1% is $81.3\times$, confirming that the top-ranked pixels are genuinely enriched for future designations even when the absolute precision is low. A consequence of this lift is that screening only the top-1% highest-risk pixels captures approximately 81.4% of all test-period US designations (recall at 1%), the relevant figure for a risk officer asking “what fraction of future PA events would I have flagged?” rather than “how

many of my flagged pixels actually transitioned?” The Random Forest achieves a marginally higher ROC-AUC than LightGBM in the US setting, confirming that the RF is a competitive alternative and that results are not sensitive to model choice.

A striking feature of the US data is the geographic concentration of designations: approximately 85% of all test-period PA establishments fall within the Deserts and Xeric Shrublands biome (Bureau of Land Management public land in the American West). This concentration simplifies the prediction problem in one sense, since any model that learns to prioritise arid BLM land will score well overall, while creating severe imbalance for all other biomes.

The LOBO analysis reveals near-zero generalisation across US biomes, with a mean LOBO PR-AUC of 0.001 across evaluable biomes. All biome folds achieve PR-AUC below 0.002, confirming that the model’s discrimination is highly biome-specific and does not transfer across ecological contexts. This is a genuine institutional finding rather than a modelling artefact: PA designation in the US is driven primarily by federal land tenure, which determines whether parcels can be administratively designated. This binary signal is not captured in the model’s geospatial feature set and does not follow the ecological gradients that enable cross-biome transfer in South America and Southeast Asia. The LOBO near-zero scores thus diagnose a real data gap — absent land-ownership covariates, cross-biome transfer is not achievable.

Per-biome LOBO results are shown in Figure 14 (Appendix F). The spatial distribution of predicted probabilities across the US is shown in Figure 18 (Appendix F). Full metrics tables for the US are provided in Appendix A.

5.4 Southeast Asia: Robustness Check

The Southeast Asia model achieves a ROC-AUC of 0.994 on the test set, with a PR-AUC of 0.919. These high values reflect the geographic concentration of PA designation in the region: new PAs are predominantly located in forested upland areas of Borneo, Sumatra, peninsular Malaysia, and the Philippines, which differ sharply from surrounding lowland agricultural zones in terms of elevation, canopy cover, and remoteness. A classifier that simply labels high-elevation, high-canopy pixels as high-risk achieves near-perfect ROC-AUC because these pixels are spatially segregated from the bulk of the unprotected land area. The

strong aggregate discrimination therefore reflects this extreme spatial segregation as much as genuine model depth: the same mechanism that produces near-perfect LOBO scores in South America’s Atacama Desert. Within the high-risk zone, however, the model is genuine: the Lift at top 1% is $94.2\times$, confirming that the top-ranked pixels are substantively enriched for future designations rather than merely capturing the upland/lowland divide. The biome-stratified LOBO metrics (Appendix A) provide further evidence of spatially heterogeneous generalisation across Southeast Asian biomes.

The LOBO analysis for Southeast Asia yields a mean PR-AUC of 0.270 across five evaluable biomes (Figure 15, Appendix F), with strong heterogeneity. Temperate Conifer Forests (PR-AUC 0.634) and Tropical Coniferous Forests (0.603) generalise well: these biomes have distinctive high-elevation, high-canopy signatures that transfer across training contexts. By contrast, Mangroves (0.007), Tropical Dry Broadleaf Forests (0.041), and Tropical Moist Broadleaf Forests (0.066) show near-random LOBO performance, indicating that designation patterns in these biomes are driven by highly localised institutional or geopolitical factors not captured by continental-scale features. Several biome folds contain too few positive test observations to support meaningful evaluation and are excluded from the mean; they do not imply poor performance, only insufficient transition events in those land types.

The spatial distribution of predicted probabilities is shown in Figure 19 (Appendix F). Full metrics tables for Southeast Asia are provided in Appendix A.

5.5 Cross-Model Performance Comparison

Figure 8 situates all six trained models (three regions $\times$ two algorithms) in a unified diagnostic framework. Panel (a) shows how targeting precision degrades as the risk zone widens from the top-1% to the top-10% of pixels. Panel (b) relates global discrimination (ROC-AUC) to practical targeting effectiveness (Lift@1%), placing both algorithms for each region in the same space.

Two findings stand out. First, Southeast Asia and South America share nearly identical *absolute* precision profiles despite very different ROC-AUC values: at the top-1% threshold, both the South American ($P=0.53$) and Southeast Asian ($P=0.56$) LightGBM models flag pixels that subsequently became protected areas with roughly equal accuracy. The United

States, by contrast, achieves dramatically lower absolute precision ($P=0.033$ at top-1%), not because discrimination is weak—ROC-AUC exceeds 0.94—but because designation events are so rare (baseline rate $\approx 0.04\%$) that even a highly ranked pixel is unlikely to transition. This exposes an important distinction: Lift measures how many times *better than random* the model performs, while Precision measures the absolute hit rate a policy-maker would observe in the field. Both are high in Southeast Asia (Lift $94\times$, $P=0.56$), but the US achieves extreme Lift ($81\times$, RF: $100\times$) against a near-zero baseline.

Second, Random Forest and LightGBM occupy distinct positions within each region (panel b). In the United States the RF achieves near-perfect ROC-AUC (0.999) and Lift@1% of $100\times$, while the LightGBM calibration advantage is more pronounced (lower Brier score). In South America the two algorithms converge, with RF slightly improving ROC-AUC (0.927 vs. 0.897) at marginal cost to calibration. These within-region differences motivate the ensemble benchmarking reported in Appendix A.

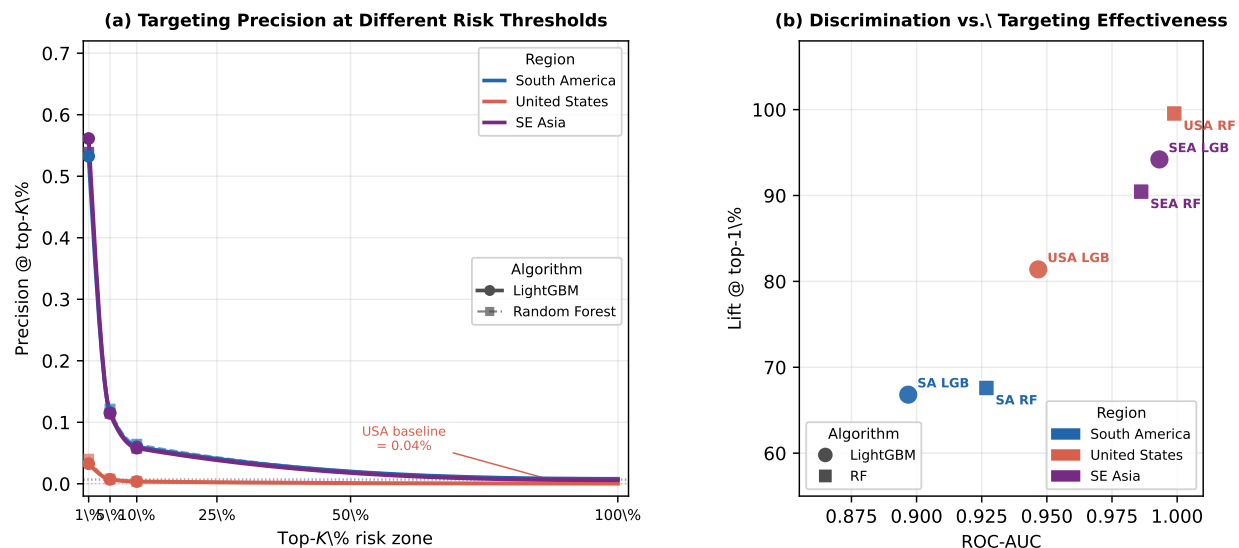


Figure 8: Cross-model performance comparison across all three regions (temporal test set 2017–2019). *Left:* Precision@K% as a function of the top-K% risk threshold for LightGBM (solid) and Random Forest (dashed). Horizontal dotted lines mark each region’s baseline positive rate. The United States curve lies far below the others because designation events are twenty times rarer than in South America or Southeast Asia, yielding near-zero absolute precision despite strong rank discrimination. *Right:* ROC-AUC vs. Lift@1% for all six model–algorithm combinations; circles denote LightGBM, squares denote Random Forest.

5.6 Cross-Regional Transfer

To test whether learned patterns are universal or continent-specific, the production models are applied across all three continental pairs without retraining, using the common 73-feature set. The SA–USA corridor provides the cleanest test: the SA LightGBM model applied to the US test set achieves ROC-AUC = 0.517; the US model applied to the SA test set achieves ROC-AUC = 0.518, both indistinguishable from a random classifier. PR-AUC is below 0.01 in both directions.

Extending the evaluation to Southeast Asia reveals a more heterogeneous but consistent pattern of transfer failure. LGBM transfers from the SE Asia model into the Americas are severely below random: SE Asia $\rightarrow$ South America ROC-AUC = 0.381; SE Asia $\rightarrow$ USA ROC-AUC = 0.238. The reverse directions are near-random (SA $\rightarrow$ SE Asia: 0.521) or moderately above (USA $\rightarrow$ SE Asia: 0.617). Below-random performance indicates that the features most predictive of designation in SE Asia are *anti-correlated* with designation in the Americas: what the SE Asia model ranks as high-risk is systematically low-risk in the other regions. Full results across all twelve directions (LightGBM and Random Forest cross-regional transfer for all three regions) are in Appendix C.

This finding of failed or inverted transfer despite an identical 73-feature set is itself a primary result, not just a model limitation. It establishes the *institutional specificity* of protected-area designation: the spatial pattern of PA creation is not derivable from physical geography alone but is instead determined by the interaction of geographic covariates with continent-level governance structures, land-tenure regimes, and conservation histories. What predicts designation in the Amazon does not predict designation in the Rocky Mountains or Borneo because the mapping from geographic features to political feasibility is entirely different across governance contexts. Even indicators of general institutional quality (WGI Rule of Law, V-Dem) cannot bridge this gap, because the specific form of institutional action (congressional monument designations in the US versus presidential decrees in the Amazon versus national park gazettement in Indonesia) is beyond what country-level composites can proxy.

Importantly, this transfer failure is not evidence that the models are wrong; it is evidence

that they work exactly as intended. Each regional model has learned the specific mapping from geographic features to designation risk that prevails in its own governance context. The fact that this mapping does not transport across continents confirms that the models capture institutionally conditioned decision-making, not merely universal biophysics.

A practical corollary is that reliable transition-risk assessments require region-specific models. A single global risk score would conflate governance contexts that produce fundamentally different feature-to-designation relationships, degrading predictive accuracy in every region.

5.7 Forward Projection: Consistency Check

Before presenting forward projections, the deployment model (trained on 2001–2019) is compared with the evaluation model (trained on 2001–2016) to verify that extending the training window does not destabilise the learned risk surface. If the two models assigned substantially different rankings to the same pixels, the forward projections would reflect training-data sensitivity rather than genuine geographic risk.

The per-pixel Pearson correlation between evaluation-model and deployment-model scores across South America is $r = 0.36$ ($n = 46.8$ million pixels). This moderate correlation confirms that risk concentrations are spatially stable across training windows: pixels the evaluation model ranked highest tend to remain high-risk under the deployment model. The correlation is not 1.0 because the two models genuinely differ in their information sets: the deployment model has three additional years of designation history, and covariate values (deforestation, PA network expansion, population) have changed between 2016 and 2019. A moderate and positive correlation is therefore the expected outcome of a well-functioning model, not a reason for concern; a correlation near zero or negative would indicate instability. The backtesting validation in Section 5.7 provides additional confirmation that the forward procedure is methodologically sound.

For the United States, the analogous correlation is $r = 0.07$ ($n = 36.7$ million pixels). This lower value reflects a statistical property of the US data rather than model instability: with a positive rate of only 0.04%, both models assign near-zero calibrated probabilities to the vast majority of pixels. Pearson correlation on a distribution concentrated near zero

is dominated by numerical differences in the tail that do not necessarily reflect different risk orderings. The evaluation model’s strong Lift@1% of $81.3\times$ and recall@1% of 81.4% confirm that the top-ranked pixels are genuinely enriched for future designations, and the spatial concentration of high-risk predictions in historically active designation areas (federal lands in the Southwest and Pacific Northwest) is consistent across both models. Forward projections for the United States are best interpreted as identifying the spatial *pattern* of transition risk rather than precise probability estimates.

5.8 Forward Projection (2025–2030)

5.8.1 On-Track Assessment

Table 4 summarises the current protection level, the business-as-usual (BAU) trajectory, and the implied gap to the 30% target for each region.

Table 4: On-track assessment for the 30×30 target. BAU trajectory extrapolates the 2019–2024 designation rate forward.

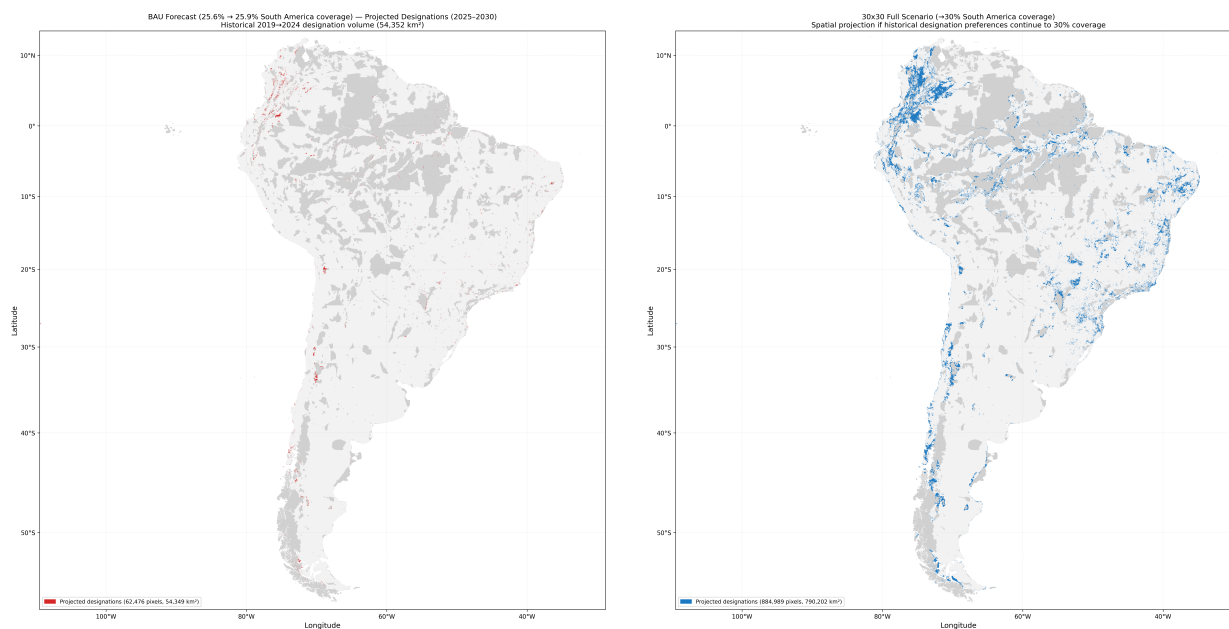
Region	Coverage	Gap to 30%	BAU rate (pct. pts./yr)	Shortfall
South America	25.6%	4.4%	0.065	~67 yr late
United States	7.7%	22.3%	<0.13	>170 yr late
Southeast Asia	14.0%	16.0%	<0.09	>170 yr late

Under the business-as-usual scenario, no region is on track. South America faces the smallest gap, having already reached 25.6% protection, but even at its recent designation pace it would not reach 30% until approximately 2093. The United States and Southeast Asia face far larger deficits relative to their target: both would require more than 170 additional years at current rates.

5.8.2 Scenario Maps and Hotspot Clusters

The forward model identifies spatially explicit priority areas under each scenario. Under the 30×30 scenario, the South America model designates the top-probability pixels until the cumulative area reaches 30% of terrestrial coverage. Spatial clustering on these highest-risk pixels identifies 40 distinct clusters in South America (Figure 9), 39 in the United States,

and 9 in Southeast Asia. The relatively small cluster count in Southeast Asia reflects the geographic structure of the region’s risk surface: high-probability pixels are concentrated in a handful of large, contiguous upland forest areas (Borneo highlands, Sumatra’s Bukit Barisan), producing a few large clusters rather than many small ones.



(a) Business-as-usual (BAU) scenario. Clusters show pixels designated if historical rates continue to 2030.

(b) 30×30 scenario. Clusters show the highest-probability pixels that would need to be designated to reach 30% terrestrial coverage by 2030.

Figure 9: Forward projection for South America under the BAU and 30×30 scenarios. Coloured clusters indicate spatially contiguous groups of high-probability pixels; grey shows the current protected-area network. Comparing the two panels illustrates the additional designation pressure required to reach the 30×30 target beyond business-as-usual trajectories.

In South America, the highest-risk clusters concentrate in the Southern Amazon (Rondônia, Mato Grosso, Pará), the Colombian Amazon, the Cerrado–Amazon transition zone, and the Andes–Amazon piedmont. These are also the areas that have seen the most active designation over 2015–2024. This convergence between predicted forward risk and recent historical patterns provides an additional out-of-sample validation of the model. Zoomed panels for the two highest-ranked South America clusters are shown in Figure 20 (Appendix F), illustrating the spatial coherence of predicted designation pressure at sub-national scale. The zoomed panels illustrate the sub-national precision of the model’s risk surface; they are provided in the appendix to keep the main text concise.

The United States clusters (Figure 16, Appendix F) concentrate in the Desert Southwest (Arizona, Nevada, Utah), the Pacific Northwest, and the Southeast, areas with large areas of Bureau of Land Management and National Forest land that historically receive new wilderness and monument designations.

Southeast Asia clusters (Figure 17, Appendix F) are concentrated in northern and central Borneo, Sumatra’s Bukit Barisan range, and the Philippines’ mountain biomes.

5.8.3 Economic Exposure in High-Risk Zones

The forward model’s risk surface is directly actionable for transition-risk analysis. To quantify the economic stakes, the area of agricultural land falling within the top-1% and top-5% predicted risk zones is computed using MODIS land-cover classifications applied to the forward-scored pixel set. Because the 1 km² cell is both the resolution limit of the covariates and the minimum unit of PA designation policy, each pixel’s risk score can be aggregated over any larger unit of interest, such as a farm boundary, a watershed, or a lending portfolio, simply by averaging or summing constituent cell scores.

In South America, 582 km² of agricultural land falls within the highest-risk 1% of unprotected pixels ($\approx$ 138,000 km² total); the low share (0.4% of the top-1% zone) reflects the dominance of remote forested and shrubland pixels in the high-risk zone, consistent with the opportunity-cost mechanism documented in Section 5.2.4: land with high agricultural value is actively suppressed from high-risk zones by the model. Widening to the top-5% zone raises the agricultural exposure to 6,143 km² (0.9% of that zone), still modest, confirming that in South America the transition-risk premium falls predominantly on natural land.

In the United States, 9,949 km² of agricultural land falls within the top-1% risk zone ($\approx$ 73,000 km² total; 13.6% agricultural share), substantially higher than South America’s 0.4%. The top-5% zone raises this to 115,620 km² (32.3%), reflecting the extensive mixture of private agricultural land and federal public land across the American West. This cross-regional contrast is substantively important: in the US, designation risk co-occurs with agricultural activity at a much higher rate, making transition exposure directly material to agricultural lenders and landowners, not only to conservation-focused investors.

In Southeast Asia, 37 km² falls within the top-1% zone ($<$ 0.1%), rising to 1,189 km²

in the top-5% zone (0.6%), again reflecting the concentration of high-risk pixels in remote forested uplands far from agricultural zones. Table 5 summarises these figures. Taken together, the consistently low agricultural shares for South America and Southeast Asia confirm the SHAP result: the model correctly learned that agricultural land value suppresses designation, so high-risk zones are dominated by natural land. The US exception, where agricultural land constitutes a substantial fraction of high-risk pixels, reflects the importance of federal land tenure in that context, where ownership (public vs. private) is a stronger predictor of designation than agricultural value. These estimates are coarse first approximations; more granular portfolio analysis would require parcel-level land-use and ownership data.

Table 5: Agricultural land exposed to designation risk in the top-1% and top-5% risk zones, by region (LightGBM forward model; MODIS land-cover). “Share” is the fraction of the respective risk zone classified as agricultural.

Region	Top-1% risk zone		Top-5% risk zone	
	Agri. area	Share	Agri. area	Share
South America	582 km ²	0.4%	6,143 km ²	0.9%
United States	9,949 km ²	13.6%	115,620 km ²	32.3%
Southeast Asia	37 km ²	<0.1%	1,189 km ²	0.6%

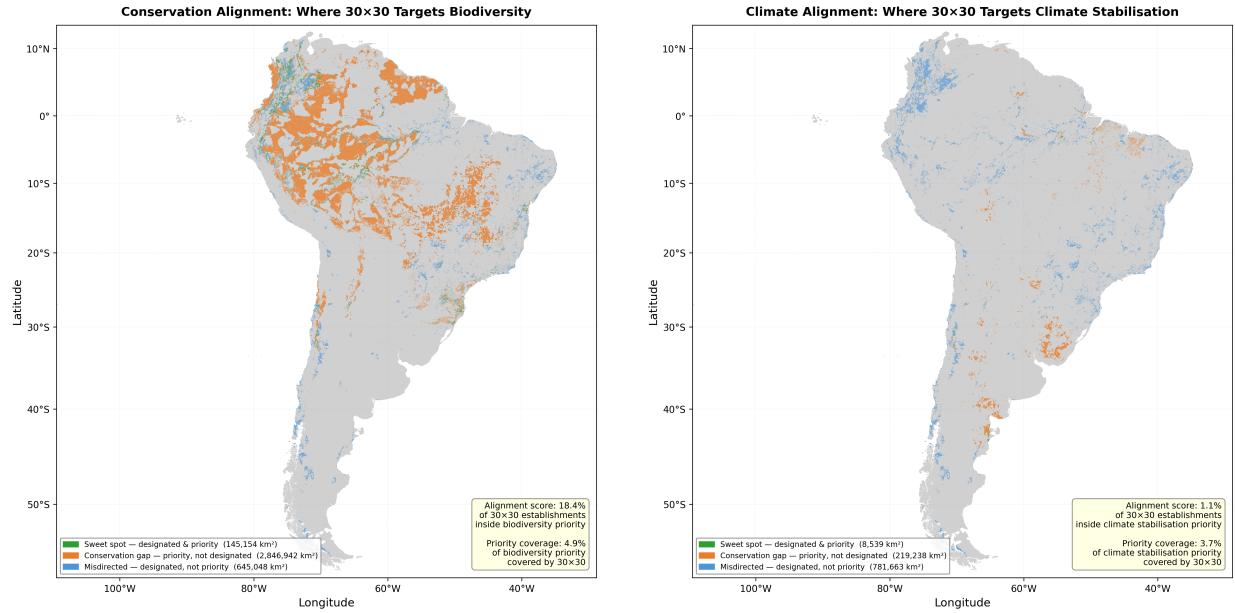
5.8.4 Overlap with Ecological Priority Areas

The representation gap — the divergence between *where governments will designate* and *where designation is most ecologically important* — is quantified by overlaying the 30×30 scenario map against two GSN priority layers: high-biodiversity areas (GSN_b2) and climate-stabilisation priority areas (GSN_b1).

Under the business-as-usual trajectory, only 18.4% of projected 30×30 designations (790,202 km² total) fall within GSN biodiversity priority zones, and these designations would cover just 4.9% of the full biodiversity priority area (2,992,096 km²). The *sweet spot* — land that is both projected for designation and classified as biodiversity-priority — amounts to 145,154 km². The overlap with climate-stabilisation priority areas is even smaller: only 1.1% of projected designations intersect these zones, covering 3.7% of the climate-stabilisation priority area (227,777 km²), with a sweet spot of 8,539 km².

Figure 10 maps these overlaps spatially. The concentration of high-risk pixels along

already-protected frontiers in the Amazon and Cerrado largely misses the interior high-biodiversity zones of the western Amazon, the Atlantic Forest remnants, and the Cerrado hotspots that remain critically underprotected. This confirms the qualitative argument in Section 6: expansion following historical patterns will not close the representation gap — it will perpetuate the bias towards remote, low-opportunity-cost land at the expense of the most ecologically irreplaceable areas.



(a) Biodiversity priority overlap

Green: GSN priority; Orange: 30×30 designations; Yellow: overlap

(b) Climate-stabilisation priority overlap

Green: GSN priority; Orange: 30×30 designations; Yellow: overlap

Figure 10: Overlap between projected 30×30 designations and GSN ecological priority areas in South America (LightGBM). Biodiversity-priority overlap: 18.4%; climate-stabilisation overlap: 1.1%. Path-of-least-resistance expansion does not target the most ecologically critical land.

6 Discussion

Driver hierarchy and the political economy of designation. The SHAP analysis reveals a driver hierarchy in which biodiversity value and governance quality are the primary predictors of designation, with economic opportunity cost entering as a secondary but substantive constraint. These are predictive associations, not causal claims: the framework cannot establish whether stronger governance causes higher designation rates, or whether both reflect deeper country-level characteristics such as income or international integration. GSN

conservation-priority indicators rank above all other variable groups; governance indicators (WGI Rule of Law, V-Dem Liberal Democracy Index) rank second, ahead of infrastructure-based proxies for economic activity. This is not the purely cost-driven “least-resistance” story that early empirical work suggested (Joppa and Pfaff, 2009; Venter et al., 2018): rather, institutional capacity to act on conservation commitments is a binding constraint of comparable importance to economic opportunity cost. The two channels are complementary rather than competing: regions combine all three conditions, ecological priority, institutional capacity, and low economic cost, for the bulk of their historical designations.

The implication for the 30×30 target is therefore twofold. First, in countries with weak institutions, even ecologically important areas may remain unprotected not because they are economically valuable but because the state lacks the administrative capacity to gazette and enforce new PAs. Second, in countries with strong governance, the remaining unprotected biodiversity-priority land tends to be economically valuable, requiring the compensation mechanisms (direct payments, conservation easements, carbon market incentives) that are central to the political economy of ambitious designation. If future designations follow the same governance-and-ecology logic as historical ones, the additional 12 percentage points of coverage will concentrate in countries with both strong institutions and available low-cost land, leaving biodiversity-rich areas in governance-constrained contexts systematically underrepresented (Venter et al., 2018; Eckert et al., 2023). The governance-biodiversity mismatch is starkest in regions such as the Congo Basin and parts of Melanesia and New Guinea, where some of the world’s highest-priority forest ecosystems sit in countries that historically have low designation rates relative to their biodiversity endowment, precisely the kind of gap that the BAU forward projections in this thesis cannot be expected to close.

The SA PR-AUC of 0.645 (against a random baseline of 0.008) is consistent with the level of discrimination achieved by Mouillot et al. (2024) using a comparable socio-environmental niche approach, providing cross-study validation that geographic and institutional covariates contain sufficient signal to predict PA designation at continental scale.

The representation gap. The LOBO analysis reveals a predictable pattern of biome underrepresentation under historical dynamics. Biomes with consistently high LOBO performance, including arid deserts and certain tropical forests, are those where designation

follows regular geographic gradients and where the economic opportunity cost of protection is low. Biomes with poor LOBO performance, such as temperate conifer forests in the US and montane grasslands in South America, exhibit idiosyncratic designation patterns driven by institutional and political factors that the model cannot capture from generic geospatial covariates alone.

This implies a representation gap: even if 30% terrestrial coverage is achieved by 2030, the distribution of new PAs across biomes is likely to remain biased toward ecologically “easy” land. Closing this gap requires active policy intervention: conservation finance instruments, direct payments to landowners, or binding biome-specific designation targets that shift designations toward high-opportunity-cost areas.

The economic exposure results (Section 5.8.3) reveal the converse dimension of this gap. In South America and Southeast Asia, the high-risk zone is overwhelmingly composed of remote forest, shrubland, and savanna: areas that are both ecologically valuable and economically cheap to designate, which is precisely why they score highly under historical designation patterns. The agricultural share of the top-1% risk zone is below 1% in both regions. This confirms that predicted designations in these regions serve a dual purpose: they are both politically feasible (low economic cost) and ecologically relevant (high biodiversity value). The designation risk for agricultural landowners is therefore concentrated at the edges of these natural landscapes, not at the core. In the United States, by contrast, the top-1% zone has a 13.6% agricultural share, reflecting the mixed public-agricultural landscape of the West where ecological and economic use coexist. Here the predicted designations are less purely “ecologically easy” and carry a higher direct cost to working agricultural businesses, which also makes them more politically contested.

Cross-continental consistency and limits of generalisability. The near-random or below-random cross-continental transfer performance (SA→USA LGBM ROC-AUC 0.517; USA→SA 0.518; SE Asia → USA 0.238; SE Asia → SA 0.381) establishes a clear boundary for the model’s generalisability, and should be read as a substantive empirical result rather than a modelling shortcoming. Protected-area designation is *institutionally specific*: its spatial pattern is not a universal law derivable from physical geography but a product of specific national legal frameworks, political histories, and institutional capacities (Ludwig et al.,

2023). Even a feature set that includes both remotely-sensed covariates and country-level governance indicators cannot capture the specific congressional acts that create a national monument in Utah or the presidential decrees that establish Amazonian reserves, because the *mapping* from general institutional quality to specific spatial designation patterns differs fundamentally across governance regimes.

This finding has two practical implications. For risk assessment, it confirms that region-specific models are required: a single global risk score would obscure continent-level variation in the drivers of designation. For research, it suggests that the most valuable next-generation features are spatially granular institutional ones: parcel-level land-tenure status (public vs. private ownership), which country-level governance indicators cannot proxy, carbon-market eligibility at the project level, and NGO funding flows that enable or constrain specific designation processes.

Calibrated probabilities as transition-risk scores. The core value proposition of this framework for financial risk applications depends on calibration. A raw ranking of pixels by predicted score is useful for conservation prioritisation, but transition-risk quantification requires statements of the form “this parcel has a 3% probability of being designated within five years”: an interval-scale probability, not an ordinal rank.

After Platt scaling, the ECE of 0.0081 in South America confirms that the calibrated scores have this property: on average, the predicted probability deviates from the observed transition rate by less than one percentage point across the score distribution. A lender extending a five-year agricultural loan secured on land in the top-1% risk zone could use the calibrated $\hat{p}$ directly to adjust the expected return, price a covenant trigger, or flag the exposure for additional due diligence, applications analogous to the use of credit default probabilities in fixed-income risk management.

The model produces a conditional forecast, not an unconditional one: probabilities are calibrated to historical designation patterns and assume that the policy environment does not change structurally. If the Kunming-Montreal commitments trigger a sustained acceleration in designation pace, calibrated probabilities would underestimate true risk. In this sense, the BAU scenario is a lower bound on transition risk rather than a point forecast.

A second-order effect worth noting for financial analysis is that large-scale designation

under 30×30 may *raise* the value of the remaining undesignated agricultural land by increasing its effective scarcity. Investors holding agricultural or agribusiness assets in regions with high designation probability face a two-sided exposure: the parcel itself may be designated (a direct transition-risk loss), but adjacent or neighbouring agricultural land that escapes designation could appreciate as usable agricultural area contracts. This scarcity premium is not modelled here; calibrated probabilities capture only the designation risk side, and practitioners using the risk maps for portfolio analysis should account for this indirect effect.

Forward projection and policy implications. The on-track assessment in Table 4 delivers a clear message: no region studied here will reach the 30×30 target on time under business-as-usual trajectories. South America, despite having the smallest percentage-point gap, would arrive approximately 67 years late; the United States and Southeast Asia face deficits of more than 170 years at current rates.

Closing this gap requires not only more designations but also different designations. The scenario maps identify which specific geographic clusters would need to be designated under each pathway. Governments committed to the 30×30 target can use these maps in two complementary ways. First, as a prioritisation tool: the high-probability clusters identify which areas are most consistent with historical designation patterns and thus most politically feasible to protect in the near term. Second, as a gap analysis: comparing the scenario maps to biodiversity priority maps (Key Biodiversity Areas, IUCN Red List habitat maps) reveals which high-priority areas are *not* predicted to be designated under business-as-usual (the representation gap) and therefore require special policy attention.

The distinction between governments that need to *accelerate* designation and those that need to *redirect* it matters. South America needs primarily to accelerate; the US needs to redirect as well as accelerate, because the existing gap is too large to fill by protecting only the historically favoured desert and shrubland landscapes.

Limitations. The model extrapolates historical designation patterns forward and is subject to a stationarity assumption. Structural breaks, such as a change in government, a new carbon-market mechanism, the Kunming-Montreal momentum, or emerging nature-related financial disclosure requirements (Taskforce on Nature-related Financial Disclosures, 2023), are not modelled. Calibrated probabilities should be interpreted as conditional forecasts:

what would the five-year designation probability be, given that the historical pattern continues? They are not unconditional predictions of what will happen. For transition-risk applications, conditional forecasts are the appropriate product; they answer the question investors and lenders actually face: what is the risk given current trajectories? The BAU scenario is therefore best read as a lower bound on designation risk, conservative in the sense that post-COP15 political momentum, if sustained, would push actual designation rates above the historical baseline the model captures.

The current feature set includes country-level governance indicators (WGI, V-Dem, DPI) and indigenous territory boundaries (LandMark), but is missing key spatially granular institutional variables: parcel-level land-tenure status (public vs. private ownership) and carbon-market project eligibility (REDD+ credits). Country-level governance quality imperfectly proxies the parcel-level tenure constraints that drive many designation decisions, particularly in mixed federal-private landscapes such as the US West. NGO funding flows, which directly enable specific designation processes, are also unobserved. Incorporating parcel-level tenure data, where available, would be the highest-priority extension.

Two data-quality issues affect the target variable. Gazette year entries in the WDPA may lag the formal legal designation by one to three years in some jurisdictions (Bingham et al., 2019); the five-year lookahead window reduces sensitivity to small date errors, but systematic lags in data entry could introduce a small bias toward later observation of transitions. More structurally, the WDPA’s increasing incorporation of Other Effective Area-Based Conservation Measures (OECMs), which vary widely in protection strength from strictly managed nature reserves to loosely governed community areas, may introduce heterogeneity in the target variable not captured by the binary protected/unprotected classification. OECMs and indigenous conserved areas often arise precisely where conventional PA designation has not been politically feasible: in areas with higher human pressure, active resource-use rights, or contested land tenure. A model trained on historical WDPA entries dominated by classical PA types would not learn the socio-environmental signature of OECM siting, and forward projections calibrated to traditional PA patterns could systematically miss the areas most likely to receive OECM designations under the Kunming-Montreal Framework.

Country-level composite indicators (WGI, V-Dem) and remotely sensed proxies (NDVI,

night-time lights) are measured with error relative to the true underlying quantities they approximate. Measurement error attenuates estimated feature importance and can reduce discrimination. Predictors measured with greater spatial precision would be expected to improve discrimination, particularly in biomes where the current LOBO performance is near-random. Relatedly, neighbouring pixels share similar covariate values and often share designation outcomes because entire landscapes are designated together. This spatial autocorrelation can inflate within-region evaluation metrics, including LOBO, because training pixels from other biomes still contain information correlated with the held-out biome through shared continental gradients in climate and terrain. The LOBO framework mitigates this by holding out entire biomes, but cannot fully eliminate it; users of LOBO metrics in spatial ecological models should interpret them as upper bounds on true out-of-sample performance.

Several methodological choices warrant sensitivity caveats. The distance-to-existing-PA feature (`dist_wdpa`) captures spatial path dependence but risks encoding a subtle endogeneity: large protected areas are sometimes designated in successive administrative steps over several years, so pixels immediately adjacent to an existing PA may score highly simply because they are part of the same multi-year designation process rather than because they are independently at risk. The one-year temporal lag on `dist_wdpa` mitigates but does not eliminate this concern. Temporal weighting likewise introduces a modelling choice: observations in later training years receive higher weight, and if the post-2010 acceleration in designation pace is a temporary policy phase rather than a persistent structural shift, this weighting may cause the model to over-extrapolate recent dynamics. Both sensitivities would require full HPC re-training runs to formally test. Additionally, road proximity and night-time lights are strong predictors partly because they proxy accessibility, but also because they correlate with economic development levels that independently influence the political feasibility of protection; the model cannot separately identify these mechanisms, and causal interpretation of the feature importance is therefore not warranted.

The relative contribution of each predictor group has been characterised through SHAP values, but a formal ablation study removing each group in turn and measuring the change in PR-AUC has not been conducted. Ablation would provide stronger evidence that the individual predictor groups independently add predictive value rather than acting as proxies

for one another, and remains a priority for future work.

Finally, pixels assigned high predicted probability that are not designated within the five-year evaluation window are classified as false positives by conventional metrics. However, spatial inspection of false positives from the walk-forward backtests (Appendix D) reveals that a substantial share cluster along known designation frontiers, in the same geographic corridors that received PAs in adjacent time windows. Many false positives are therefore better characterised as “right prediction, wrong window”: the model correctly identifies them as elevated-risk zones, but the designation is delayed beyond the five-year horizon. This is an inherent constraint of the fixed-window binary target rather than a model failure. The persistence of false positive clusters across independent backtest origins (2009, 2011) provides indirect evidence that the forward risk maps are spatially consistent: the same frontier zones recur as high-risk across all periods evaluated, increasing confidence that the 2025–2030 forward predictions identify genuine long-run designation candidates rather than noise. Throughout, the framework is explicitly predictive, not causal: SHAP values should be read as partial correlations, not causal effects, and the appropriate use of this framework is spatial risk screening and portfolio prioritisation, not policy impact evaluation.

7 Conclusion

This thesis has developed and evaluated a machine-learning framework for predicting the spatial expansion of protected areas under the 30×30 biodiversity target, applied to three major world regions. The primary model for South America achieves a ROC-AUC of 0.897 and captures 90.4% of all test-period PA establishments within the top 1% of predicted risk, performance sufficient to make the model a practically useful tool for spatial transition-risk screening. Calibrated predicted probabilities serve as interval-scale transition-risk scores that can be used directly in financial risk quantification, portfolio screening, and land-use planning.

Three empirical findings emerge from the analysis. The driver hierarchy reveals that conservation priority (GSN biodiversity indicators) and governance quality (WGI Rule of Law, V-Dem Liberal Democracy Index) are the primary predictors of where PAs are des-

ignated, not merely economic opportunity cost. If future designations continue to follow the same governance-and-ecology logic as historical ones, closing the 30×30 gap will require strengthening institutions and providing compensation mechanisms in biodiversity-rich, economically contested areas, not only designating remote land. The gap between what this model predicts and where designation *should* go for maximum conservation impact is itself a policy finding: it reveals where the 30×30 commitment risks being met on paper without delivering proportionate biodiversity outcomes. Forward projections reinforce the urgency: all three studied regions are severely off-track, South America by approximately 67 years and the United States and Southeast Asia by more than 170 years at current designation rates. Perhaps the most striking result is that cross-continental transfer performance is near-random or below-random across all evaluated directions, confirming that designation patterns are products of specific governance histories and institutional contexts rather than universal ecological gradients, and that region-specific models are required for credible risk assessment.

For governments, the most actionable output is the scenario maps: they identify specific geographic clusters that would need to be designated under BAU, moderate, and full 30×30 scenarios, and can be compared against biodiversity priority maps to distinguish between acceleration-only and acceleration-plus-redirection policy needs. For investors and lenders, the calibrated probability surface provides a readily deployable transition-risk score for any land parcel in the three regions studied. For researchers, the governance-as-primary-driver finding motivates models that add parcel-level land tenure (public vs. private), carbon-market project eligibility, and NGO funding flows, the spatially granular institutional predictors that country-level governance scores cannot proxy.

The most important extension is the inclusion of parcel-level land-tenure data. Even though the current model includes country-level governance indicators, parcel-level public versus private ownership is qualitatively different in its direct effect on political feasibility, and its absence is the most likely proximate cause of cross-continental transfer failure. A second valuable extension is to update the calibration annually as new WDPA data become available, maintaining probability scores that reflect the current pace of policy implementation rather than historical averages.

Three methodological extensions would further strengthen the framework. First, a formal ablation study, removing each predictor group (biodiversity, governance, economic value, climate, path-dependence) in turn and measuring the change in PR-AUC, would provide stronger evidence for the driver hierarchy than SHAP values alone, and would quantify the incremental value of each data source for practitioners deciding where to invest in data acquisition. Second, replacing the uniform spatial smoothing kernels with a learnable spatial encoder such as a U-Net architecture would allow the model to identify multi-scale context features without the need to pre-specify kernel sizes, and could capture irregular landscape geometries (river corridors, mountain ridges) that fixed-size kernels miss. Third, the pipeline structure is region-agnostic: the same preprocessing, training, and evaluation stages could be extended to Tropical Africa and Temperate Asia with minimal re-engineering, allowing a global coverage probability surface aligned with the full 30×30 ambition.

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Appendix

A Extended Results: LOBO and Regional Evaluation Tables

This appendix consolidates the per-biome Leave-One-Biome-Out (LOBO) PR-AUC tables and the full evaluation metrics for the United States and Southeast Asia models. Tables 6, 7, and 8 report per-biome LOBO PR-AUC for all three regions (LightGBM models); Tables 9 and 10 report the benchmark metrics for USA and SE Asia respectively.

Table 6: South America: Per-biome LOBO PR-AUC (LightGBM). Mean = 0.216.

Biome	N positives	Pos. rate	LOBO PR-AUC
Deserts & Xeric Shrublands	109,678	10.4%	0.980
Flooded Grasslands & Savannas	1,425	2.7%	0.219
Mangroves	1,457	0.2%	0.215
Mediterranean Forests, Woodlands & Scrub	442	0.8%	0.037
Montane Grasslands & Shrublands	5,184	0.1%	0.002
Temperate Broadleaf & Mixed Forests	889	2.0%	0.035
Tropical & Subtropical Dry Broadleaf Forests	10,328	0.4%	0.232
Tropical & Subtropical Grasslands, Savannas	4,184	0.4%	0.118
Tropical & Subtropical Moist Broadleaf	62,359	0.5%	0.107
Mean			0.216

Table 7: United States: Per-biome LOBO PR-AUC (LightGBM). Mean = 0.001 (evaluated biomes only). Four biomes had zero test-period designations and could not be evaluated (N/A).

Biome	N positives	Pos. rate	LOBO PR-AUC
Deserts & Xeric Shrublands	12,625	0.21%	0.001
Mediterranean Forests, Woodlands & Scrub	216	0.11%	0.001
Temperate Broadleaf & Mixed Forests	564	0.06%	0.001
Temperate Conifer Forests	737	0.06%	0.001
Temperate Grasslands, Savannas & Shrublands	363	0.03%	0.001
Flooded Grasslands; Mangroves; Trop. Conifers; Trop. Grasslands	0	—	N/A
Mean (evaluated)			0.001

Table 8: Southeast Asia: Per-biome LOBO PR-AUC (LightGBM). Mean = 0.270 (evaluated biomes only). Three biomes had zero test-period designations.

Biome	N positives	Pos. rate	LOBO PR-AUC
Mangroves	46	—	0.007
Temperate Conifer Forests	4,284	—	0.634
Tropical & Subtropical Coniferous Forests	21	—	0.603
Tropical & Subtropical Dry Broadleaf Forests	17,702	—	0.041
Tropical & Subtropical Moist Broadleaf Forests	51,475	—	0.066
Montane Grasslands; Temperate Broadleaf; Trop. Grasslands	0	—	N/A
Mean (evaluated)			0.270

Regional Evaluation Tables

Table 9: United States: LightGBM predictive performance on the temporal test set (2017–2019). Baseline positive rate: $\approx 0.040\%$.

Metric	LightGBM
ROC-AUC	0.947
PR-AUC	0.093
Precision @ top 1%	0.033
Lift @ top 1%	81.3 $\times$
Precision @ top 5%	0.007
Lift @ top 5%	17.9 $\times$
Brier score	0.000400
ECE (post-calibration)	0.000310

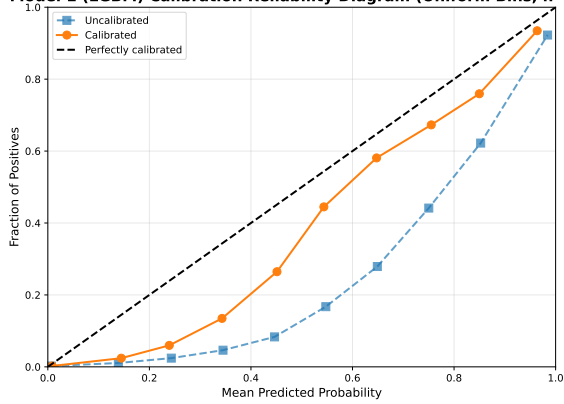
Table 10: Southeast Asia: LightGBM predictive performance on the temporal test set (2017–2019). Baseline positive rate: $\approx 0.596\%$.

Metric	LightGBM
ROC-AUC	0.994
PR-AUC	0.919
Precision @ top 1%	0.561
Lift @ top 1%	94.2 $\times$
Precision @ top 5%	0.115
Lift @ top 5%	19.3 $\times$
Brier score	0.00144
ECE (post-calibration)	0.00863

B Reliability Diagrams

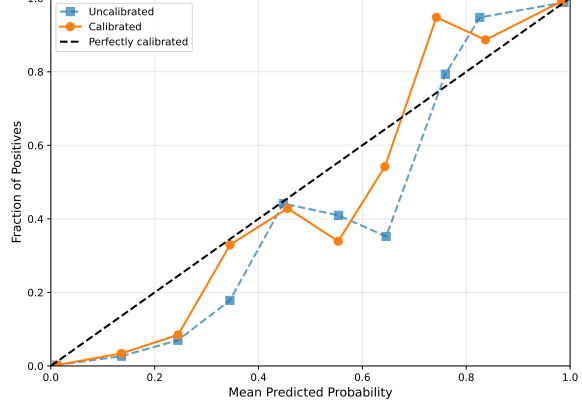
Reliability diagrams show the fraction of positive outcomes (observed transition rate) binned by predicted probability, both before and after Platt scaling (Platt, 1999). Post-calibration probabilities are well-aligned with observed rates across all regions, with ECE values below 0.011 for all LightGBM models.

Model 1 (LGBM) Calibration Reliability Diagram (Uniform Bins, n=10)



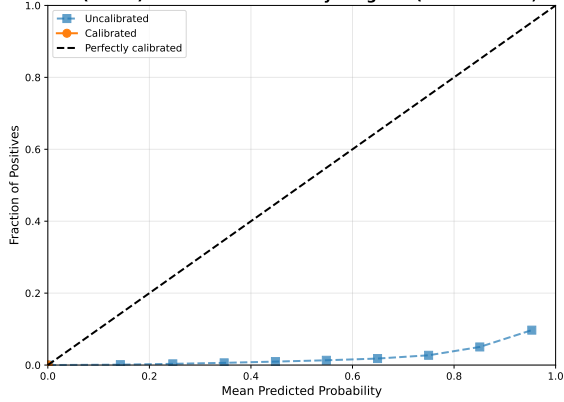
(a) South America: LightGBM

Model 1 (RF) Calibration Reliability Diagram (Uniform Bins, n=10)



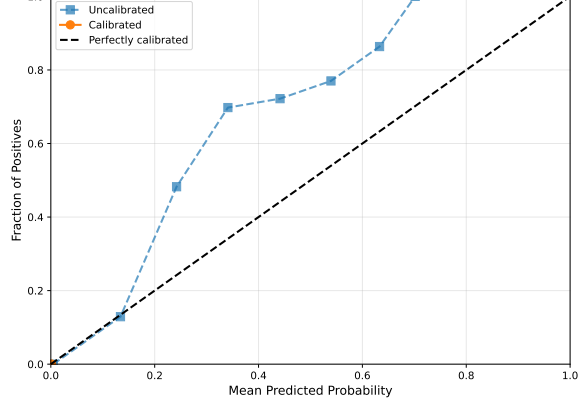
(b) South America: Random Forest

Model 2 (LGBM) Calibration Reliability Diagram (Uniform Bins, n=10)



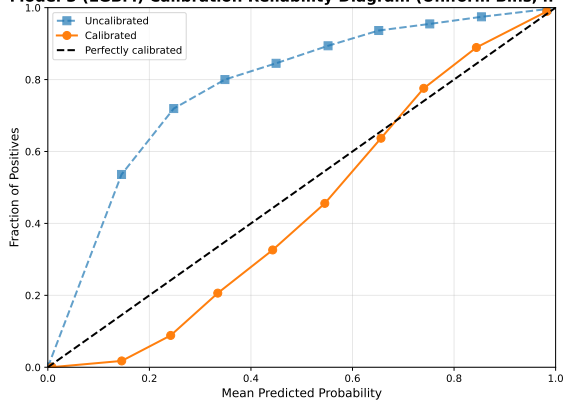
(c) United States: LightGBM

Model 2 (RF) Calibration Reliability Diagram (Uniform Bins, n=10)



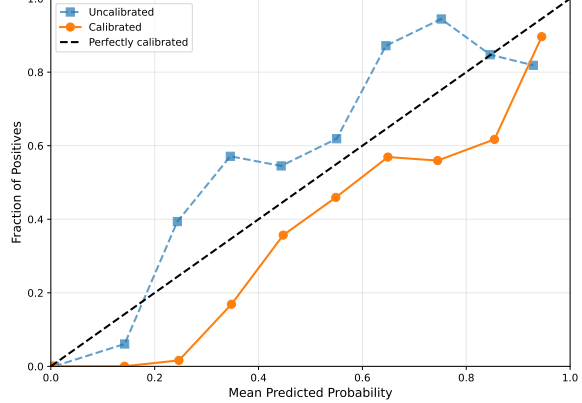
(d) United States: Random Forest

Model 3 (LGBM) Calibration Reliability Diagram (Uniform Bins, n=10)



(e) Southeast Asia: LightGBM

Model 3 (RF) Calibration Reliability Diagram (Uniform Bins, n=10)



(f) Southeast Asia: Random Forest

Figure 11: Reliability diagrams for all six model variants (before and after Platt scaling). Each bar shows the observed positive rate within the corresponding predicted-probability bin; perfect calibration is the 45° diagonal.

C Cross-Continental Transfer Detail

Cross-continental transfer is evaluated across all three continental pairs using the 73-feature set common to all regions. Each production model is serialised after training on its home region (2001–2019) and scored on each other region’s test panel (2017–2019) without any retraining. Table 11 reports ROC-AUC for all twelve direction–algorithm combinations.

Table 11: Cross-continental transfer ROC-AUC (all directions). Test years 2017–2019. Random-classifier baseline = 0.500. Values below 0.500 indicate the model has learned patterns that are anti-correlated with designation in the target region.

Direction	Algorithm	ROC-AUC	Interpretation
SA → USA	LGBM	0.517	near-random
USA → SA	LGBM	0.518	near-random
SA → SE Asia	LGBM	0.521	near-random
USA → SE Asia	LGBM	0.617	moderate transfer
SE Asia → SA	LGBM	0.381	below random
SE Asia → USA	LGBM	0.238	strongly below random
SA → USA	RF	0.301	below random
USA → SA	RF	0.531	near-random
SA → SE Asia	RF	0.720	substantial transfer
USA → SE Asia	RF	0.614	moderate transfer
SE Asia → SA	RF	0.598	moderate transfer
SE Asia → USA	RF	0.713	substantial transfer

The LGBM results are internally consistent: SA–USA LGBM transfer is near-random in both directions, while transfers involving SE Asia diverge; the SE Asia model applies anti-predictively to the Americas (ROC-AUC well below 0.5), reflecting that the feature combinations most predictive of PA designation in SE Asia (forested upland areas in Borneo and the Mekong region) are not where designation happens in the Americas. The RF results show a different pattern: SA→SE Asia and SE Asia→USA both achieve ROC-AUC ≈ 0.71 –0.72, suggesting that the bagged ensemble has learned more robust ecological baselines that partially transfer, whereas the gradient-boosted LGBM has overfit to institutional patterns specific to each region. The SA→USA RF (0.301) replicates the below-random phenomenon seen for LGBM SE Asia transfers, pointing to genuine anti-correlation between SA high-designation features and US high-designation pixels.

Taken together, the results confirm that LGBM-based transition-risk scores require region-

specific calibration, while also suggesting that future generalisation efforts may benefit from ensemble or bagging architectures that suppress institutional overfitting.

D Backtesting Results

Walk-forward backtests at origins $T \in \{2009, 2011\}$ verify that deployment-style inference generalises to out-of-sample designation events. Each origin trains on 2001–($T - 1$), scores unprotected pixels in year T , and evaluates against realised designations in years $T + 1$ through $T + 5$. SA LightGBM backtest precision@1% is stable across origins; USA precision is lower and more variable, consistent with the test-set PR-AUC.

False positive spatial patterns. Figures 13a–13b map the spatial distribution of false positives at backtest origin $T = 2017$ for South America and the United States: pixels that fall within the top-1% predicted risk zone but were not designated within the subsequent five-year window. In South America, false positives concentrate along the southern and eastern Amazon frontier (Mato Grosso, Pará) and in Andean foothills, the same zones where genuine designations are clustered, suggesting that many false positives are temporally rather than geographically misplaced: the model correctly identifies active designation frontiers but the specific parcels flagged were designated slightly before or after the evaluation window. In the United States, false positives are concentrated in arid BLM landscapes of the West, reflecting that the model assigns high risk to a large contiguous landscape zone, only a fraction of which receives formal designation in any five-year period.

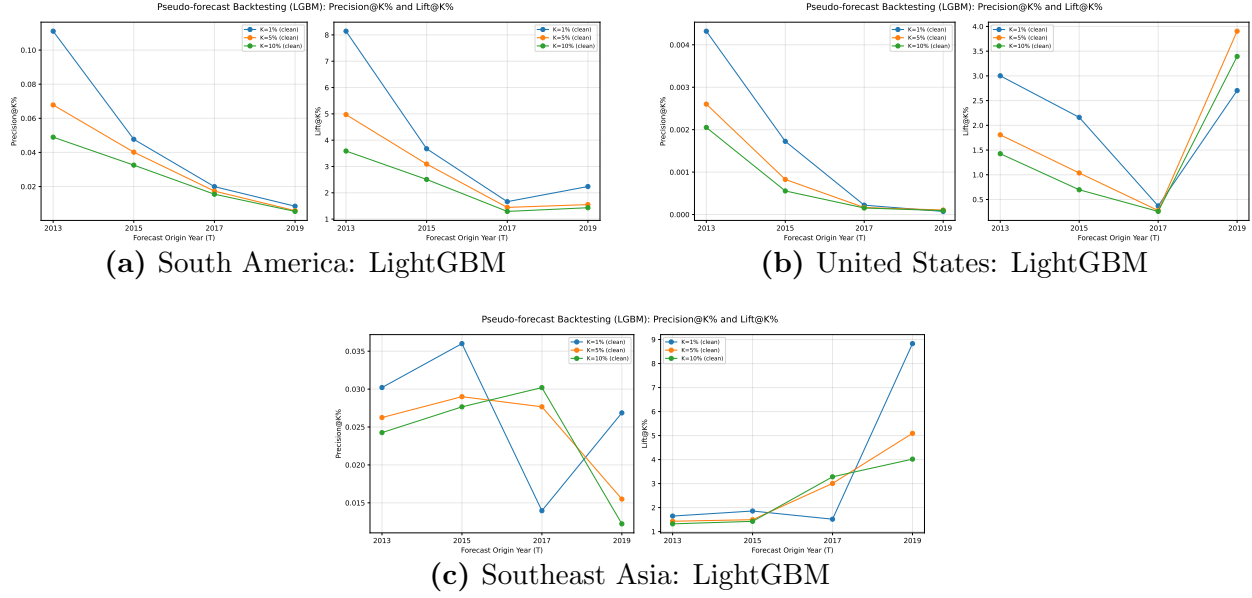


Figure 12: Backtesting precision@1% over time for walk-forward forecasts at origins $T \in \{2009, 2011\}$. Each line shows the precision of the top-1% of unprotected pixels scored in year T (i.e. the $\approx 1\%$ highest-ranked pixels among all pixels eligible in that origin year), evaluated against realised 5-year designation outcomes in years $T+1$ through $T+5$.

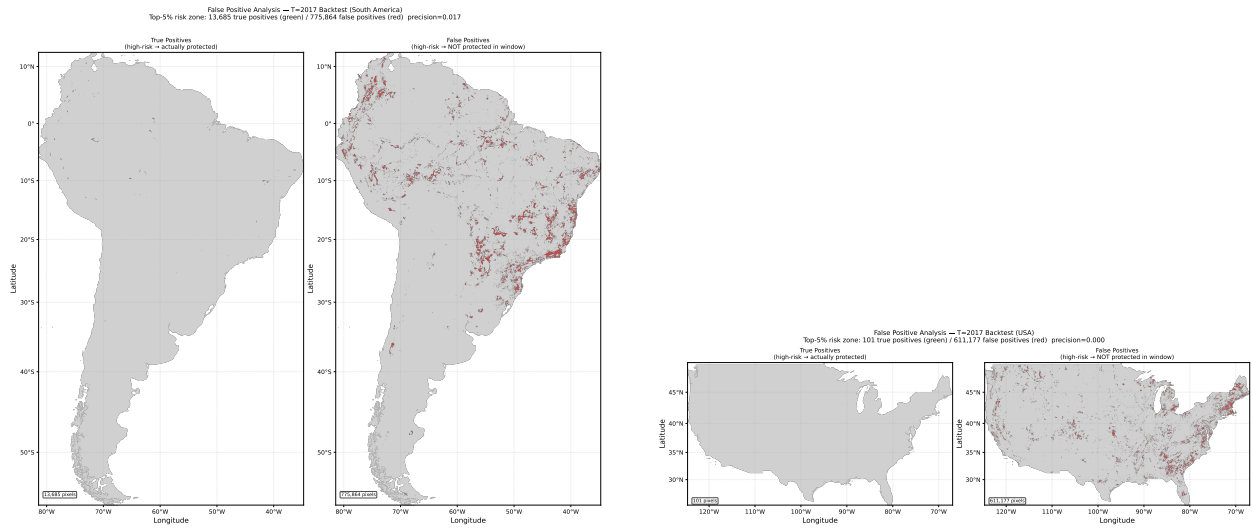


Figure 13: Spatial distribution of false positives at backtest origin $T = 2017$: pixels in the top-1% predicted risk zone that were not designated within the subsequent five-year window. Clustering of false positives in known designation frontiers (Amazon arc, BLM arid West) suggests temporal rather than geographic misclassification for most cases.

E Feature Dictionary

Table 12 lists the full feature set used in the South America, USA, and Southeast Asia models. All three regions share the same 73-feature schema, derived from the predictor groups in Table 1 via multi-scale spatial smoothing (4×4 , 16×16 , 64×64 km), Euclidean distance transforms, and temporal lags.

Table 12: Feature dictionary (selected entries). Full schema has 73 features after spatial smoothing. *Static* = time-invariant; *Dynamic* = updated annually.

Feature name	Type	Description
dist_wdpa	Dynamic	Euclidean distance to nearest PA boundary (km), lagged 1 yr
elevation_b1	Static	Elevation (m, CGIAR SRTM V4)
slope_b1	Static	Slope (degrees, derived from SRTM)
NDVI_b1	Dynamic	Annual median NDVI (MODIS MOD13Q1)
NDVI_b1_smooth4	Dynamic	NDVI averaged over 4×4 km window
NDVI_b1_smooth16	Dynamic	NDVI averaged over 16×16 km window
NDVI_b1_smooth64	Dynamic	NDVI averaged over 64×64 km window
deforestation_b1	Dynamic	Annual forest-loss indicator (Hansen)
deforestation_b1_smooth64	Dynamic	Forest loss averaged over 64×64 km window
wildfire_b1	Dynamic	Annual burned-area indicator (MODIS MCD64A1)
GPW_b1	Dynamic	Population density (GPW v4, forward-filled)
GPW_b1_smooth16	Dynamic	Population density at 16×16 km
GPW_b1_smooth64	Dynamic	Population density at 64×64 km
HNTL_b1	Dynamic	Night-time lights (HNTL, DMSP/VIIRS harmonised)
HNTL_b1_smooth64	Dynamic	Night-time lights at 64×64 km
dist_road	Static	Distance to nearest major road (km, GRIP)
dist_powerplant	Static	Distance to nearest power plant (km, WRI)

Feature name	Type	Description
dist_oil_gas	Static	Distance to nearest oil/gas infrastructure (km, Omara et al.)
dist_indigenous	Static	Distance to nearest indigenous/community land (km, LandMark)
landcover_b1	Dynamic	Annual dominant land-cover class (MODIS MCD12Q1)
economic_value_lag1	Dynamic	Agricultural land value (USD/ha, FAO FAOSTAT), lagged 1 yr
policy_b2	Dynamic	V-Dem Liberal Democracy Index (country level)
policy_b3	Dynamic	WGI Government Effectiveness (country level)
policy_b4	Dynamic	WGI Rule of Law (country level)
GSN_b2	Static	GSN high-biodiversity area indicator
GSN_b2_smooth16	Static	GSN biodiversity averaged over 16×16 km
GSN_b2_smooth64	Static	GSN biodiversity averaged over 64×64 km
WorldClim_b7	Static	WorldClim V1 annual temperature range
WorldClim_b14	Static	WorldClim V1 precipitation of driest month
WorldClim_b17	Static	WorldClim V1 precipitation seasonality
...	...	Additional WorldClim, GSN, smoothed, and policy variants (73 total)

F USA and Southeast Asia: Supplementary Figures

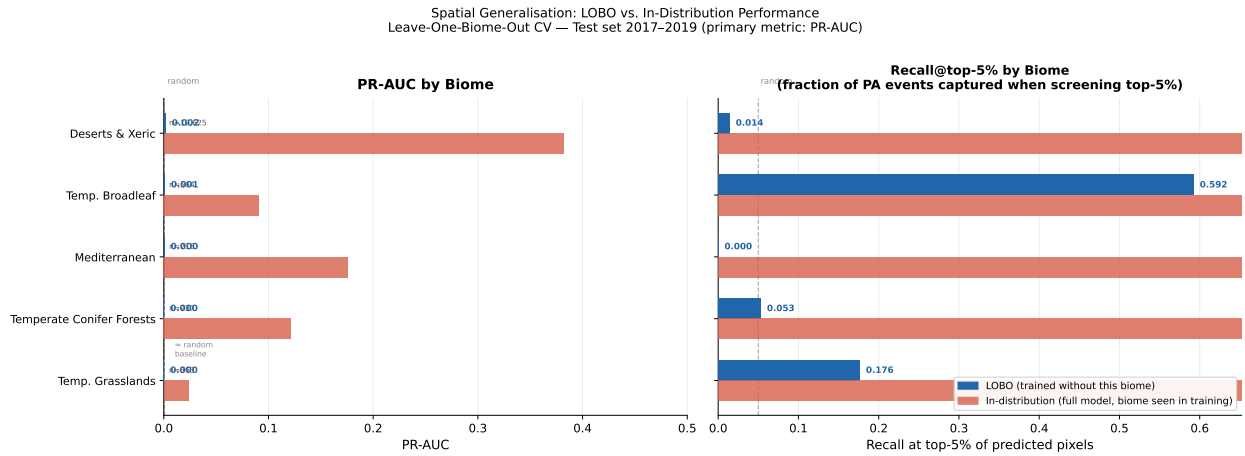


Figure 14: Per-biome LOBO PR-AUC for the United States (LightGBM, test set 2017–2019). Biomes with zero test-period designations are excluded. LOBO PR-AUC values are near zero across all biomes, reflecting that the USA model relies on federal public-land tenure patterns that do not transfer across biome boundaries.

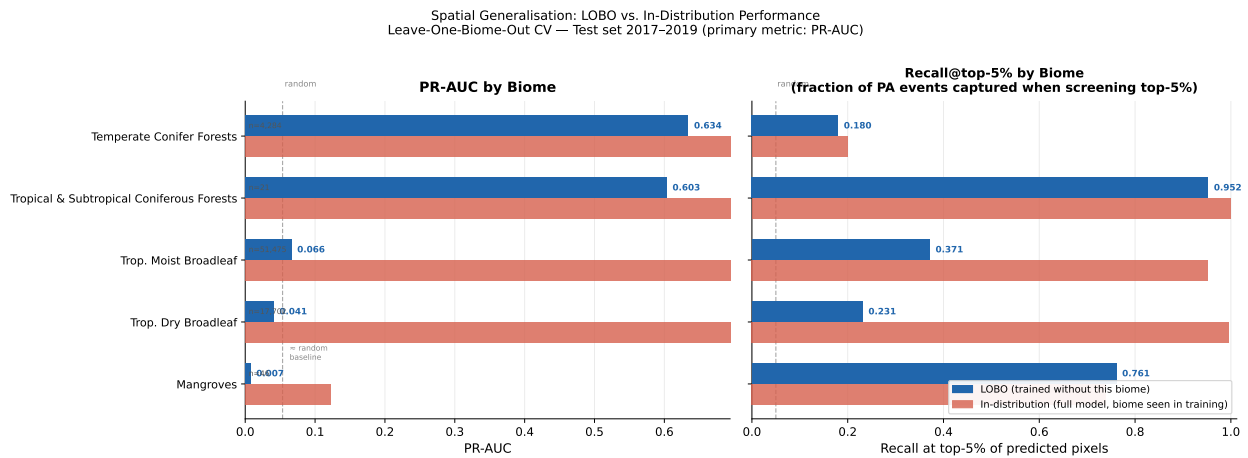
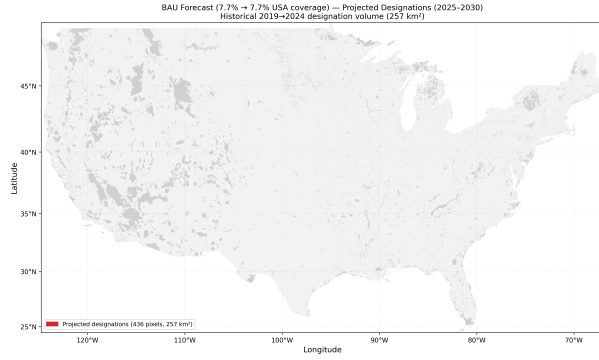
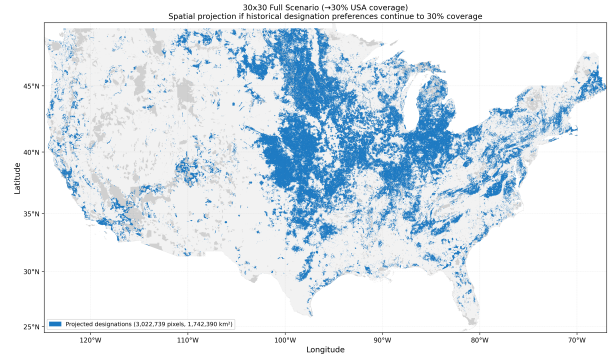


Figure 15: Per-biome LOBO PR-AUC for Southeast Asia (LightGBM, test set 2017–2019). Biomes with zero test-period designations are excluded. The mean LOBO PR-AUC is 0.169, indicating that pan-regional ecological gradients provide moderate out-of-distribution generalisation across biome types in the region.

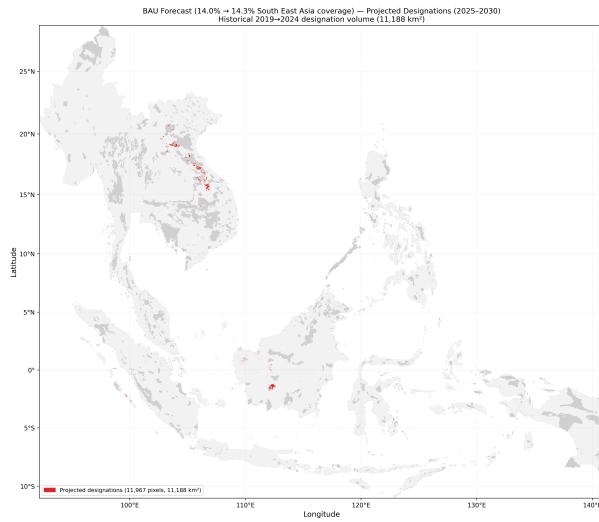


(a) BAU scenario.

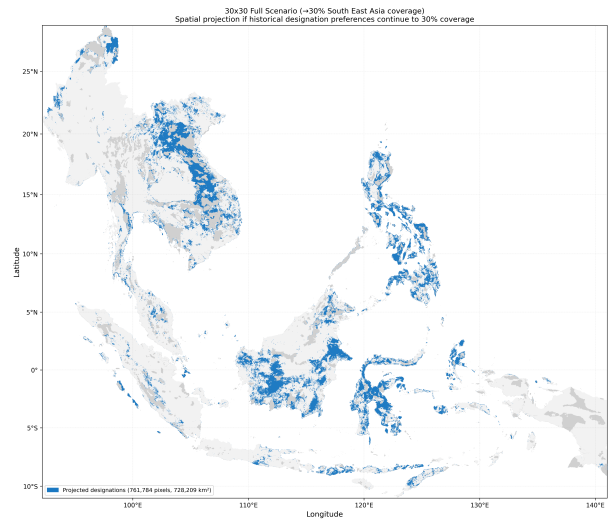


(b) 30×30 scenario.

Figure 16: Forward projection for the United States under BAU and 30×30 scenarios. Coloured clusters indicate high-probability pixels; grey shows the current protected-area network. Designation concentrates in the arid West (BLM and National Forest lands) and the Pacific Northwest.



(a) BAU scenario.



(b) 30×30 scenario.

Figure 17: Forward projection for Southeast Asia under BAU and 30×30 scenarios. The limited number of clusters (9 total) reflects the concentration of high-probability pixels in large forested upland areas in Borneo and Sumatra.

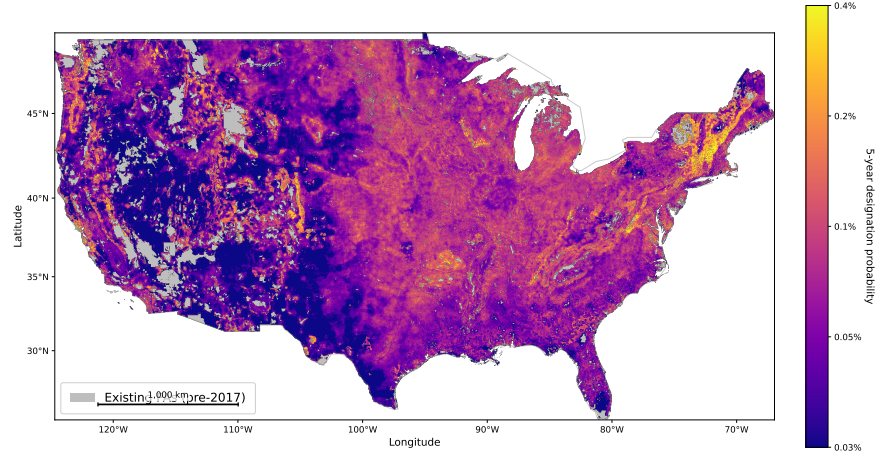


Figure 18: Predicted 5-year PA designation probability for the United States (LightGBM, deployment model). Higher probabilities concentrate in the arid West where Bureau of Land Management public land dominates. Note the compressed probability scale relative to South America, reflecting the 20× lower positive rate ($\approx 0.04\%$).

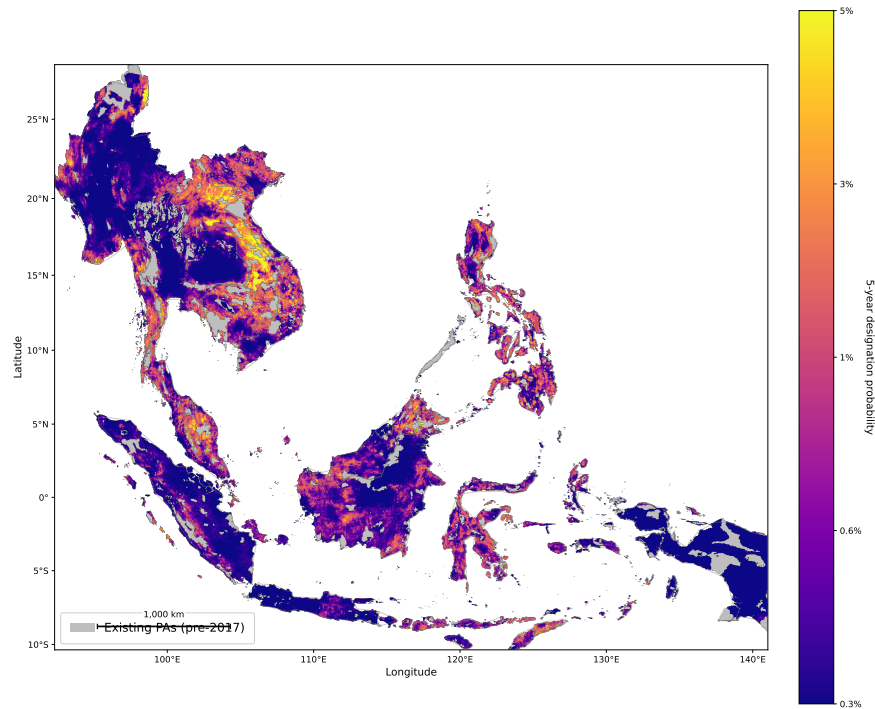


Figure 19: Predicted 5-year PA designation probability for Southeast Asia (LightGBM, deployment model). High-probability pixels cluster in the forested uplands of Borneo, Sumatra, and peninsular Malaysia, reflecting the strong elevation and canopy-cover signal in this region.

Hotspot Zoom-ins — South America
 Top row: P(protection by 2030) | Bottom row: BAU designation zone (green = existing PAs)

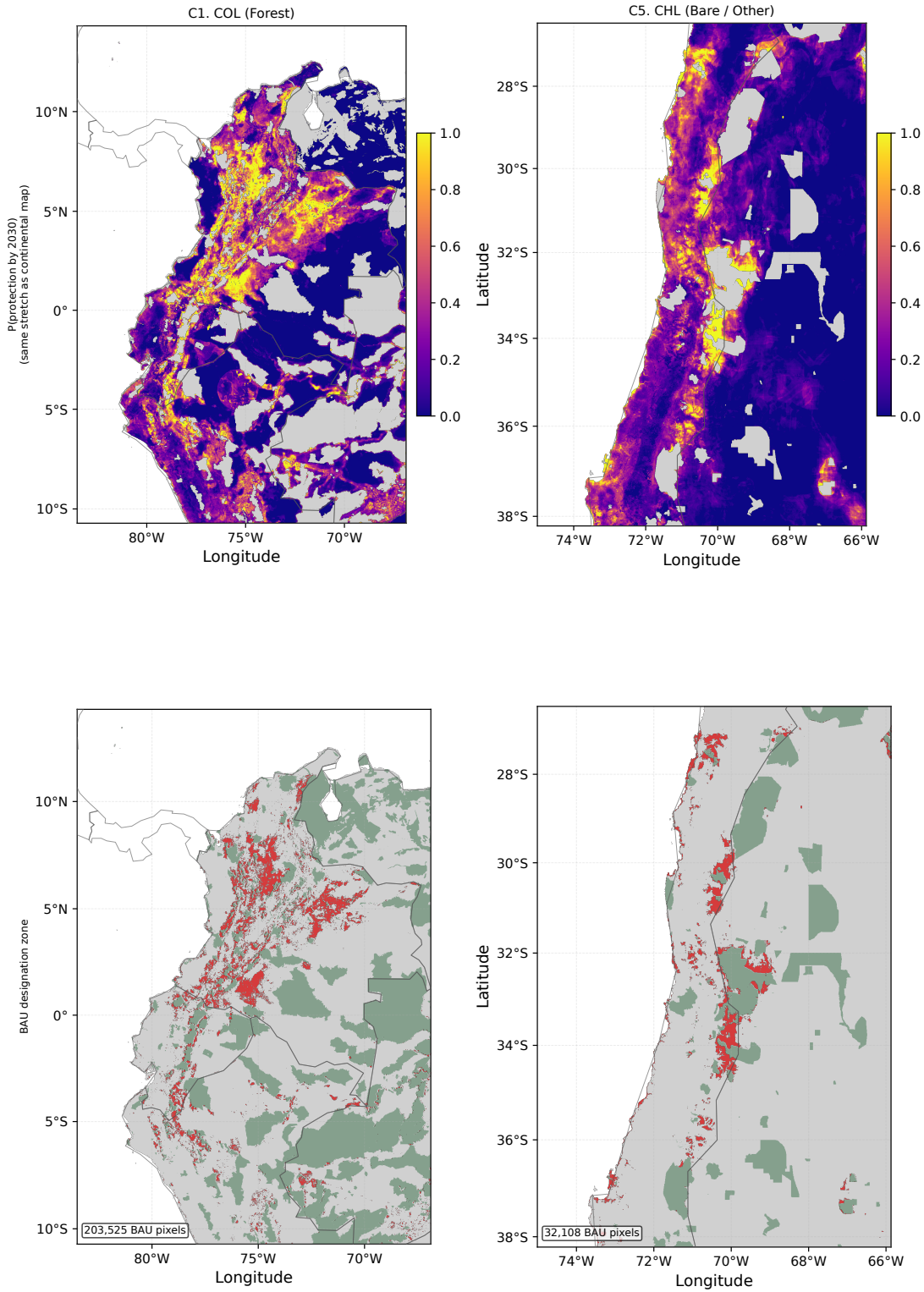


Figure 20: Hotspot zoom-ins for South America (LightGBM, BAU scenario). Top row: predicted designation probability by 2030. Bottom row: BAU designation zone (red), existing PAs in green, country borders in grey. Left: Colombian Amazon cluster (C1, highest-ranked BAU hotspot). Right: central Chilean cluster (C5, Andean foothills Bare/Other land-cover regime).

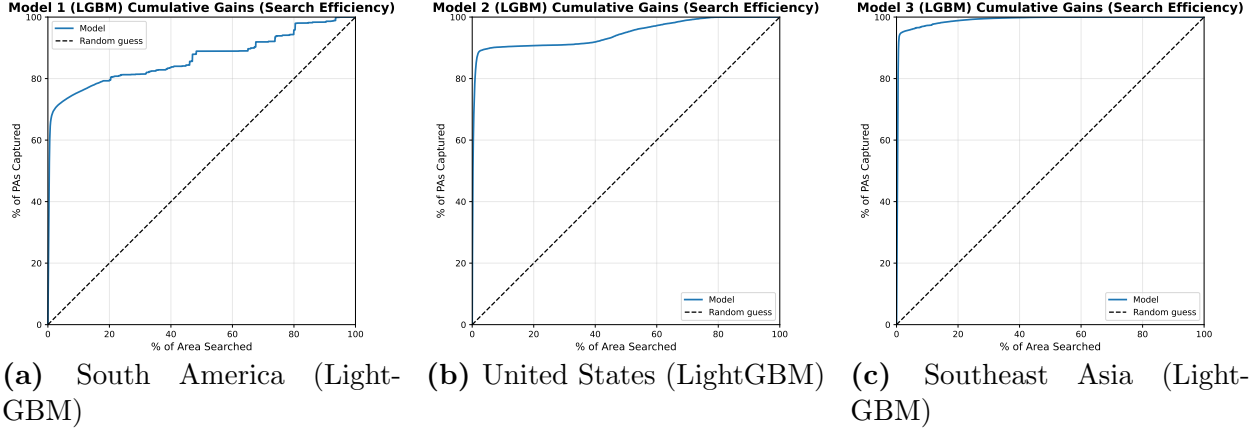


Figure 21: Cumulative gains charts for all three regions (LightGBM, test period 2017–2019). Each curve shows the fraction of all pixel-year positive observations captured as a function of the fraction of pixels scored, ranked by predicted probability. The dashed diagonal represents a random scorer (no skill).

G SHAP Dependence Plots

SHAP dependence plots show how a single feature’s value affects its SHAP contribution to the model’s log-odds output. Each point is one observation from the SHAP sample; the x -axis is the raw feature value, the y -axis is the SHAP value (positive = increases predicted designation probability, negative = decreases it). The colour of each point encodes the SHAP value of the most strongly interacting secondary feature, revealing joint effects. The following plots illustrate the three features with the most interpretively distinctive marginal effects for the South America LightGBM model.

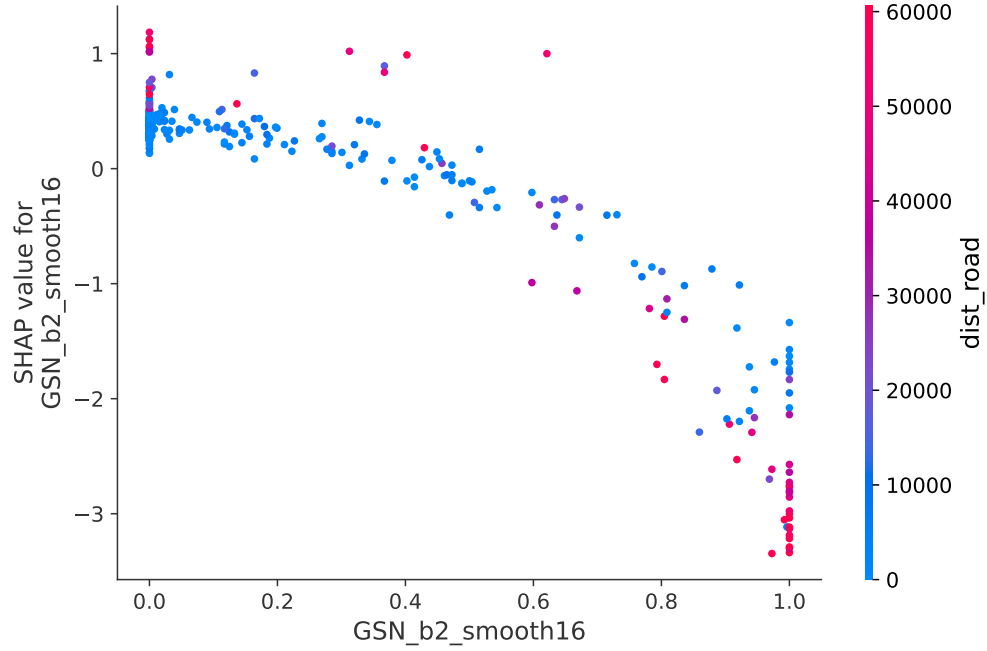


Figure 22: SHAP dependence plot for `GSN_b2_smooth16` (GSN high-biodiversity indicator, 16×16 km neighbourhood mean). The positive relationship confirms that pixels in globally significant biodiversity areas consistently attract designation regardless of other covariate values. Colour indicates the SHAP value of a secondary interacting feature.

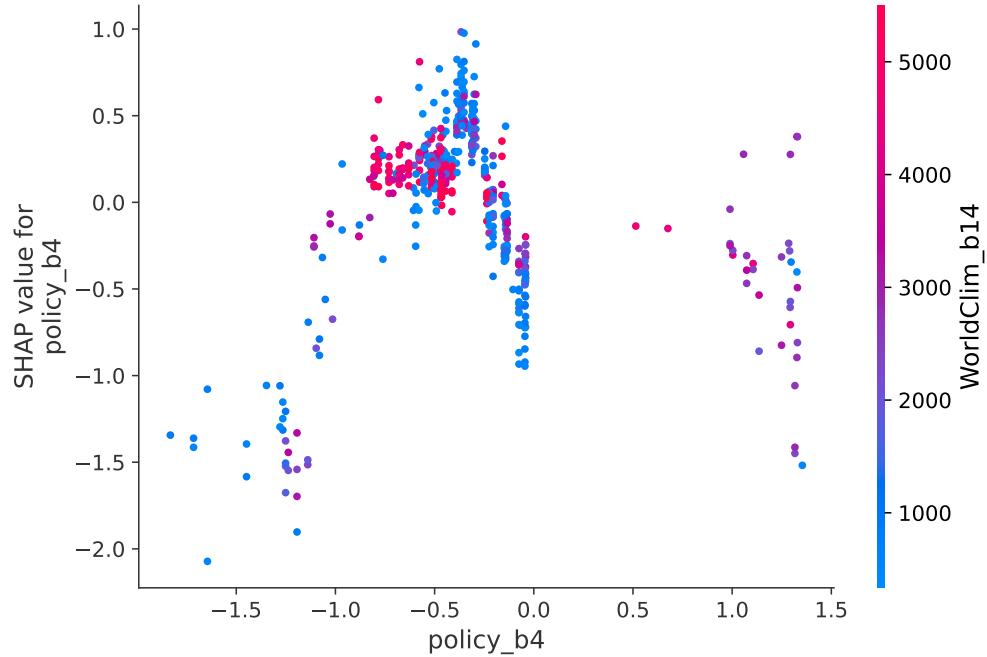


Figure 23: SHAP dependence plot for `policy_b4` (WGI Rule of Law, country level). The monotone positive relationship indicates that stronger rule-of-law institutions are associated with higher designation probability across the continent, confirming governance quality as an independent predictor rather than a proxy for ecological value.

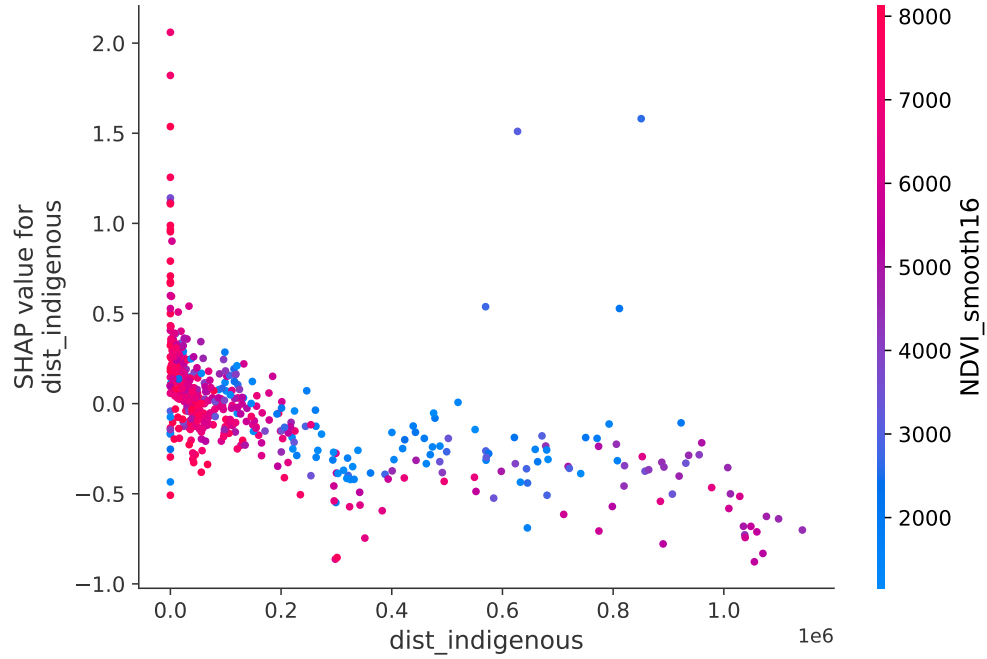


Figure 24: SHAP dependence plot for `dist_indigenous` (distance to nearest indigenous or community-managed land, km). Proximity to indigenous territories is associated with higher designation probability, consistent with co-management pathways and the overlap between indigenous territories and high-biodiversity areas in the Amazon.

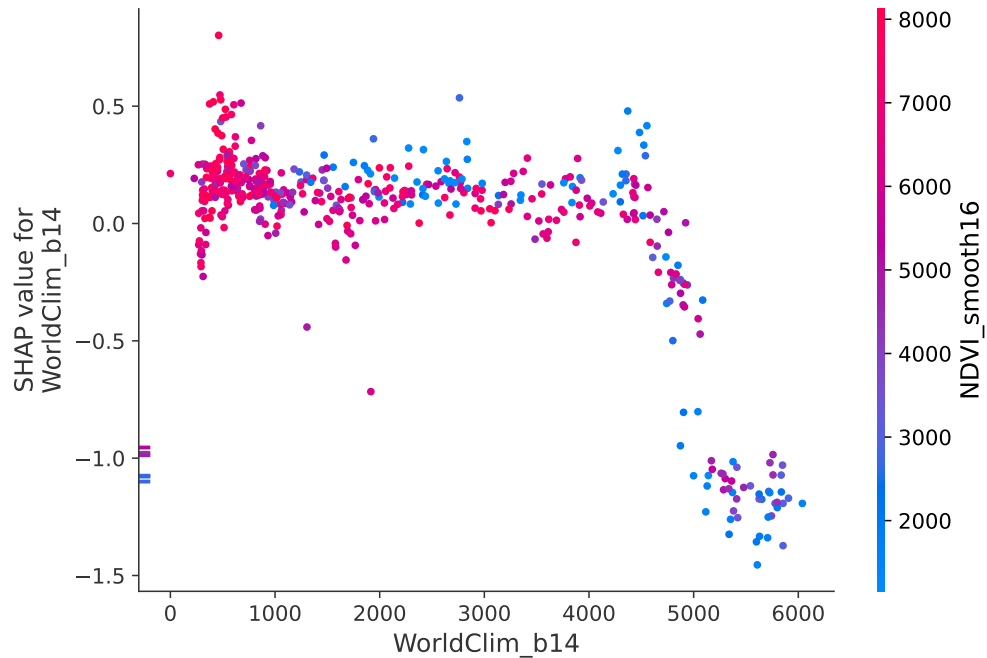


Figure 25: SHAP dependence plot for `WorldClim_b14` (precipitation of the driest month, mm). The positive SHAP at low and high extremes reflects the model's use of climate envelope information: both hyper-arid and wet-season-dominated regimes exhibit distinctive spatial designation patterns.

H Colombia Sub-Region Results

A fully separate pipeline was implemented for Colombia as a computational testbed during early model development. On the Colombia test set (2017–2019), the LightGBM model achieves ROC-AUC 0.946 and PR-AUC 0.470; the Random Forest achieves ROC-AUC 0.931 and PR-AUC 0.435. These results confirm that both model classes perform consistently at smaller geographic scales and that the continental South America model is not simply benefiting from its larger sample size.

These sub-national results provide an additional scale validation: both model classes perform consistently at the country level, confirming that the continental South America model’s discrimination is not simply a product of its larger sample size or the averaging of heterogeneous sub-regions. The Colombia results suggest that the framework generalises from continental to national scales without loss of predictive quality.